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A linearly implicit shock capturing scheme for compressible two-phase flows at all Mach numbers

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ABSTRACT

We present a semi-implicit solver for the solution of compressible two-phase flows governed by the Baer–Nunziato model. A novel linearly implicit discretization is proposed for the pressure fluxes as well as for the relaxation source terms, whereas an explicit scheme is retained for the nonlinear convective contributions. Consequently, the CFL-type stability condition on the maximum admissible time step is based only on the mean flow velocity and not on the sound speed of each phase, so that the novel scheme works uniformly for all Mach numbers. Central finite difference operators on Cartesian grids are adopted for the implicit terms, thus avoiding any need of numerical diffusion that might destroy accuracy in the low Mach number regime. To comply with high Mach number flows, shock capturing finite volume schemes are employed for the approximation of the convective fluxes. The discretization of the non-conservative terms ensures the preservation of moving equilibrium solutions, making the new method well-balanced. The new scheme is also proven to be asymptotic preserving in the low Mach limit of the mixture model. Second order of accuracy is achieved by means of an implicit-explicit (IMEX) time stepping algorithm combined with a total variation diminishing (TVD) reconstruction technique. The novel method is benchmarked against a set of test cases involving different Mach number regimes, permitting to validate both accuracy and robustness.

1. Introduction

Several natural processes are concerned with multi-phase flows, which are mathematically described by nonlinear systems of time-dependent hyperbolic partial differential equations. Relevant examples are avalanches, meteorological flows with cloud formation, volcanic eruptions, sediment transport in rivers, and debris flows in mountain regions. On the other hand, also many industrial applications involve multi-phase flows, mainly focused on combustion processes encountered in aerospace engineering, automotive industry, nuclear reactor safety, paper and food manufacturing, as well as turbo-machinery for renewable energy production. Depending on the Mach number, which measures the ratio between the fluid velocity and the sound speed, the behavior of the flow may be quite disparate, ranging from the compressible regime for high Mach numbers to the incompressible regime in the low Mach number limit. Indeed, for industrial processes, compressible phases can be typically considered as they describe the combustion of liquid and solid fuels or solid and bio-mass. However, for natural phenomena, one phase may exhibit a compressible behavior, e.g. air, while the other phase lies in the incompressible regime, e.g. water.

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Due to the complexity of such flows, no universally agreed model exists in the literature. Nevertheless, most of the models are based on the Baer–Nunziato (BN) equations for compressible two-phase flows, originally introduced for detonation waves in solid–gas combustion processes [1]. In this case, the exchange terms of mass transfer and convective heat are discarded, and each phase is governed by its own equation of state. The resulting hyperbolic system is composed by balance laws for the mass, momentum, and energy of each phase, and the evolution of the volume fraction. In addition, some source terms account for the interaction between the phases, due to friction forces and pressure. These equations allow each phase to have its own velocity and pressure. Simplified versions may be obtained making hypotheses on the problem at hand. For example, an asymptotic reduction of the model in the limit of the velocity, the pressure, or the temperature equilibrium yields a six equation model [2–6]. If both velocity and pressure are common to the two phases, the system further simplifies to five equations [7–11], and if velocity, pressure, and also temperature undergo instantaneous equilibrium, then the model becomes a four-equations one [12]. However, these models represent a subset of the complete system, and their simplified nature often comes with additional modeling difficulties [13,14]. We then concentrate on the full seven-equation Baer–Nunziato model.

The numerical solution of multi-phase flows is very challenging due to the need of treating different Mach number regimes that may coexist in the same computation. Simulating flows with different orders of magnitude of the Mach number is not trivial. Furthermore, the BN model involves non-conservative terms accounting for the interaction between the inter-phase pressure and the gradient of the volume fraction, whose discretization must ensure physical consistency. For compressible multi-phase flows, shock capturing finite volume schemes are often employed [4,15–20], that resolves complex structures developing when the Mach number is of order one [18,20,21]. Such methods are normally explicit. However, for very low Mach numbers, the effect of numerical viscosity on the slow convective waves introduced by upwind-type schemes is proven to degrade the accuracy [22,23]. Moreover, the CFL-type stability condition dictates a severe restriction on the maximum admissible time step while approaching the low Mach limit, hence making an explicit scheme no longer useful in practice. Indeed, in the incompressible or weakly compressible limit, the acoustic waves are negligible, so there is no need to supply the scheme with a large amount of numerical viscosity, that must be counterbalanced by a smaller time step and a higher mesh resolution. In this case, implicit methods are instead preferred, although they imply the solution of a globally nonlinear system defined on the entire computational domain. A successful idea consists in treating implicitly only one part of the system to be solved, while keeping the remaining explicit, thus both incompressible and compressible regimes can be handled [24–29]. This approach permits to design space and time discretizations in which the implicit part of the system is relatively easy to be inverted, typically avoiding nonlinear systems, while keeping robustness and shock-capturing properties in the explicit part. This family of scheme is typically referred to as IMplicit-EXplicit (IMEX) methods [30–37], which are proven to be asymptotic preserving, meaning that they recover a consistent discretization of the incompressible model in the low Mach number limit. A slightly different approach to design all Mach solvers, i.e. numerical schemes that can uniformly deal with a large spectrum of Mach number regimes, is given by semi-implicit methods [38–45] that aim at obtaining a linearly implicit scheme for the stiff terms in the governing equations, thus avoiding any need of iterative methods.

Although developing all Mach solvers is an active research topic, most of the numerical methods are addressed to single-phase flows, as the aforementioned works. Only few contributions may be found, in the literature, for two-phase flows. In [46], a pressure-based formulation of the Baer–Nunziato model is proposed, to simulate one-dimensional weakly-compressible flows with generic equation of state. Instead of solving for the total energy, a non-conservative evolution equation for the pressure is formulated and discretized, so that a more physically relevant set of variables is adopted. The method is thus adapted for low Mach regimes, but it lacks accuracy in the high Mach regime, especially for the resolution of shock waves, which are no longer captured [47]. The asymptotic behavior of the scheme is demonstrated by considering a unified reference Mach number for both phases, and the method is first order accurate in space and time. In [48], a two-dimensional pressure-based solver is developed for low Mach flows. In this approach, a binary factor, which might be set either to unity or to zero, switches the flow behavior from compressible to incompressible regime. Other two-phase solvers have been proposed in Varsakelis and Papalexandris [49,50], where the incompressible or compressible modeling is set *a priori* for each phase. These latter can be regarded as numerical methods for mixed models rather than as asymptotic preserving schemes. Very recently, an asymptotic preserving IMEX method for the full Baer–Nunziato model in multiple space dimensions has been presented in Malusà and Alaia [51]. The Mach number is defined separately for the two phases, to account for potentially very different regimes.

In this work, we aim at designing a semi-implicit all Mach solver for the BN model on three-dimensional Cartesian grids. Central finite difference operators are used for treating the implicit terms, while we resort to an explicit finite volume method to deal with shock waves induced by the nonlinear convective terms. Particular attention is devoted to ensure the asymptotic preserving property of the scheme in the low Mach number limit for the mixture. The resulting scheme is second order accurate in space and time for a wide range of Mach numbers. Furthermore, the scheme is required to be well-balanced, thus moving stationary equilibria of the flow are exactly maintained. As pointed out in Re and Abgrall [46], this is the physical guidance to construct physically consistent discretizations of the non-conservative terms related to the gradient of the volume fraction. Differently from Malusà and Alaia [51], the volume fraction equation will be treated implicitly, and a linearly implicit solver is devised by suitable time linearization techniques borrowed from previous existing works for single-phase flows [43]. The relaxation source terms are solved implicitly while being explicitly computable via a source splitting technique, which can achieve second order of accuracy by means of semi-implicit time integrators [41]. Consequently, the relaxation limits of the model induced by the stiff source terms does not pose any problem from the numerical viewpoint. We do not resort to any predictor-corrector strategy as in Malusà and Alaia [51], and the novel scheme exactly recovers the asymptotic preserving method forwarded in Boscheri and Pareschi [43] in the case of single-phase flow. The proposed methodology is intended as a simple, but not trivial, discretization for multi-phase flows, which can become highly complex, as in the case of volcanic applications [52]. Although such flows may involve challenging relaxation terms, strongly interacting phases, and very different regimes, they share the same homogeneous and hyperbolic structure of the Baer–Nunziato model. Therefore, this work stands as the groundwork for potentially highly complex real-world multi-phase applications tailored to the simulation of magma flows.

The rest of the paper is organized as follows. Section 2 introduces the governing equations and their non-dimensional formulation, for a clear understanding of the potential sources of stiffness due to different Mach number regimes. This analysis defines the basis for the development of the numerical scheme, which is described in Section 3. Section 4 provides a variety of test cases that demonstrate the accuracy and the robustness of the new method, while facing different flow regimes. Finally, in Section 5, some conclusions are drawn, and an outline to future developments is forwarded.

2. Mathematical model

Let $\Omega \in \mathbb{R}^d$ be a bounded domain of dimension $d \in \{1,2,3\}$, defined by a spatial position vector $\mathbf{x} = (x, y, z) \in \mathbb{R}^d$, and let $t \in \mathbb{R}_+$ denote the temporal variable. The system of governing equations is given by the compressible two-phase flow Baer–Nunziato model written in the form derived by Saurel and Abgrall [15]:

$$\begin{aligned}
&\frac{\partial}{\partial t}(\rho_f\phi_f) + \nabla\cdot(\rho_f\phi_f\mathbf{u}_f) = 0,\\
&\frac{\partial}{\partial t}(\rho_f\phi_f\mathbf{u}_f) + \nabla\cdot(\rho_f\phi_f\mathbf{u}_f\mathbf{u}_f) + \nabla(p_f\phi_f) + p_I\nabla\phi_f = \delta(\mathbf{u}_s-\mathbf{u}_f),\\
&\frac{\partial}{\partial t}(\rho_f\phi_f E_f) + \nabla\cdot(\phi_f\mathbf{u}_f(\rho_f E_f + p_f)) + p_I\mathbf{u}_I\nabla\phi_f = \mu\, p_I\,(p_s-p_f) + \delta\,\mathbf{u}_I\,(\mathbf{u}_s-\mathbf{u}_f),\\
&\frac{\partial}{\partial t}(\rho_s\phi_s) + \nabla\cdot(\rho_s\phi_s\mathbf{u}_s) = 0,\\
&\frac{\partial}{\partial t}(\rho_s\phi_s\mathbf{u}_s) + \nabla\cdot(\rho_s\phi_s\mathbf{u}_s\mathbf{u}_s) + \nabla(p_s\phi_s) - p_I\nabla\phi_s = -\delta(\mathbf{u}_s-\mathbf{u}_f),\\
&\frac{\partial}{\partial t}(\rho_s\phi_s E_s) + \nabla\cdot(\phi_s\mathbf{u}_s(\rho_s E_s + p_s)) - p_I\mathbf{u}_I\nabla\phi_s = -\mu\, p_I\,(p_s-p_f) - \delta\,\mathbf{u}_I\,(\mathbf{u}_s-\mathbf{u}_f),\\
&\frac{\partial}{\partial t}\phi_s + \mathbf{u}_I\cdot\nabla\phi_s = \mu(p_s-p_f).
\end{aligned} \tag{1}$$

System (1) represents a non-equilibrium model, where each phase $\alpha = \{f, s\}$, with f the fluid phase and s the solid phase, has its own density ρ_α, pressure p_α, velocity $\mathbf{u}_\alpha$, volume fraction ϕ_α, and specific total energy E_α. The volume fraction is a non-dimensional variable, bounded in the interval $[0,1]$, and the sum of the volume fractions must be equal to unity for consistency, here $\phi_s + \phi_f = 1$. The specific total energy is defined as the sum of the specific internal energy e_α, and the specific kinetic energy k_α: $E_\alpha = e_\alpha + k_\alpha$, with $k_\alpha = \frac{1}{2}|\mathbf{u}_\alpha|^2$. In this paper, we consider the stiffened gas equation of state (EOS) for each phase:

$$e_\alpha = \frac{p_\alpha + \gamma_\alpha\pi_\alpha}{\rho_\alpha(\gamma_\alpha - 1)}, \tag{2}$$

with γ_α and π_α being, respectively, the polytropic index, and the pressure constant of each phase. The ideal gas EOS can be retrieved from (2) by setting $\pi_\alpha = 0$. The source terms in (1) account for inter-phase drag and pressure relaxation through the parameters δ and μ, respectively, which might become stiff. Indeed, the dynamics of wave propagation is typically significantly slower than the relaxation of pressure and velocity, to the point that they are frequently treated as instantaneous processes, with infinite μ and δ. The interface pressure and velocity, respectively p_I and $\mathbf{u}_I$, may be defined in different ways [53], and here we consider two definitions:

$$p_I = p_f, \qquad \mathbf{u}_I = \mathbf{u}_s, \tag{3a}$$

$$p_I = \sum_\alpha \phi_\alpha p_\alpha, \qquad \mathbf{u}_I = \frac{\sum_\alpha \rho_\alpha\phi_\alpha\mathbf{u}_\alpha}{\sum_\alpha \rho_\alpha\phi_\alpha}. \tag{3b}$$

The first definition (3a), based on a physical viewpoint, is adopted in the original Baer–Nunziato model [1], as well as in other works [54,55]. However, many authors [13,15,46,51] prefer the weighted formulation (3b). Here, we allow for both definitions, and the selection is problem-dependent. Such a choice does not affect the numerical method, but it may influence the results. The system of equations tends to the single-phase Euler equations far from the interfaces, and when considering either $\phi_s = 1$ or $\phi_f = 1$. As analyzed in Andrianov and Warnecke [56], the eigenvalues of the Baer–Nunziato model, projected in the normal direction, write

$$\lambda = \left[\mathbf{u}_f\cdot\mathbf{n},\ \mathbf{u}_f\cdot\mathbf{n} \pm c_f,\ \mathbf{u}_s\cdot\mathbf{n},\ \mathbf{u}_s\cdot\mathbf{n} \pm c_s,\ \mathbf{u}_I\cdot\mathbf{n}\right], \tag{4}$$

with $\mathbf{n}$ denoting a unit vector pointing normal to the flow direction, and $c_\alpha = \sqrt{\gamma_\alpha(p_\alpha + \pi_\alpha)/\rho_\alpha}$ representing the speed of sound of the different phases.

The Baer–Nunziato system (1) may be cast in the following general non-conservative form:

$$\frac{\partial\mathbf{Q}}{\partial t} + \nabla\cdot\mathbf{F}(\mathbf{Q}) + \mathbf{B}\cdot\nabla\mathbf{Q} = \mathbf{S}(\mathbf{Q}), \qquad \mathbf{x}\in\Omega\subset\mathbb{R}^d, \quad t\in\mathbb{R}_+, \tag{5}$$

where $\mathbf{Q} = (\rho_f\phi_f, \rho_f\phi_f\mathbf{u}_f, \rho_f\phi_f E_f, \rho_s\phi_s, \rho_s\phi_s\mathbf{u}_s, \rho_s\phi_s E_s, \phi_s)^\top$ is the state vector, $\mathbf{F}(\mathbf{Q})$ is the conservative nonlinear flux tensor, the matrix $\mathbf{B}(\mathbf{Q})$ accounts for the non-conservative contributions, and the vector $\mathbf{S}(\mathbf{Q})$ contains the source terms.

If the equations are discretized by means of a fully explicit scheme, the time step Δt is constrained by a CFL-type stability condition, that is

$$\Delta t \leq \mathrm{CFL}\min_{\Omega}\left(\frac{\varrho}{\max(|\mathbf{u}_\alpha\cdot\mathbf{n}|\pm c_\alpha)}\right), \tag{6}$$

where ϱ is the characteristic mesh size of the control volumes used to pave the computational domain. For very slow dynamics, i.e. in the low Mach regime, the sound speed increases to infinity, and Δt becomes prohibitively small. Furthermore, an excessive numerical dissipation in explicit schemes causes a loss of accuracy, for low Mach number flows. To overcome these issues simultaneously, the terms that are affected by the Mach number have to be discretized implicitly.

2.1. Non-dimensional form of the Baer–Nunziato model

To identify those terms which are responsible for the stiffness of the problem in the low Mach regime, the non-dimensional form of the equations is needed. To this purpose, we define non-dimensional quantities, characterized by a tilde symbol, scaled by reference values, with the subscript "0". Assuming that the convective scales of the two phases are comparable, we can use reference values common to both phases, hence:

$$\tilde{\mathbf{x}}_\alpha=\frac{\mathbf{x}_\alpha}{L_0},\quad \tilde{\mathbf{u}}_\alpha=\frac{\mathbf{u}_\alpha}{\mathbf{u}_0},\quad \tilde{t}=\frac{t}{\tau_0}=\frac{|\mathbf{u}_0|}{L_0}t,\quad \tilde{\rho}_\alpha=\frac{\rho_\alpha}{\rho_0},\quad \tilde{p}_\alpha=\frac{p_\alpha}{p_0},$$
$$\tilde{E}_\alpha=\frac{\rho_0}{p_0}E_\alpha,\quad \tilde{\delta}=\frac{\delta}{\delta_0}=\frac{\tau_0}{\rho_0}\delta,\quad \tilde{\mu}=\frac{\mu}{\mu_0}=\rho_0\mathbf{u}_0^2\tau_0\mu.$$

The volume fraction ϕ_α is inherently dimensionless. The scaling of the time variable holds when considering the Strouhal number, $St=\frac{L_0}{\tau_0|\mathbf{u}_0|}$, assumed to be $St\approx 1$, as usually done in the literature [57], meaning that the characteristic time scale of the flow field evolution corresponds to that one related to the convection.

The non-dimensional form of the Baer–Nunziato system is then given by

$$\begin{aligned}
&\frac{\rho_0}{\tau_0}\frac{\partial}{\partial\tilde{t}}(\tilde{\rho}_f\tilde{\phi}_f)+\frac{\rho_0\mathbf{u}_0}{L_0}\nabla\cdot(\tilde{\rho}_f\tilde{\phi}_f\tilde{\mathbf{u}}_f)=0,\\
&\frac{\rho_0\mathbf{u}_0}{\tau_0}\frac{\partial}{\partial\tilde{t}}(\tilde{\rho}_f\tilde{\phi}_f\tilde{\mathbf{u}}_f)+\frac{\rho_0\mathbf{u}_0^2}{L_0}\nabla\cdot(\tilde{\rho}_f\tilde{\phi}_f\tilde{\mathbf{u}}_f\tilde{\mathbf{u}}_f)+\frac{p_0}{L_0}\nabla(\tilde{p}_f\tilde{\phi}_f)+\frac{p_0}{L_0}\tilde{p}_I\nabla\tilde{\phi}_f=\frac{\rho_0\mathbf{u}_0}{\tau_0}\tilde{\delta}(\tilde{\mathbf{u}}_s-\tilde{\mathbf{u}}_f),\\
&\frac{p_0}{\tau_0}\frac{\partial}{\partial\tilde{t}}(\tilde{\rho}_f\tilde{\phi}_f\tilde{E}_f)+\frac{p_0\mathbf{u}_0}{L_0}\nabla\cdot(\tilde{\phi}_f\tilde{\mathbf{u}}_f(\tilde{\rho}_f\tilde{E}_f+\tilde{p}_f))+\frac{p_0\mathbf{u}_0}{L_0}\tilde{p}_I\tilde{\mathbf{u}}_I\nabla\tilde{\phi}_f=\frac{p_0^2}{\rho_0\mathbf{u}_0^2\tau_0}\tilde{\mu}\,\tilde{p}_I\,(\tilde{p}_s-\tilde{p}_f)+\frac{\rho_0\mathbf{u}_0^2}{\tau_0}\tilde{\delta}\,\tilde{\mathbf{u}}_I\,(\tilde{\mathbf{u}}_s-\tilde{\mathbf{u}}_f),\\
&\frac{\rho_0}{\tau_0}\frac{\partial}{\partial\tilde{t}}(\tilde{\rho}_s\tilde{\phi}_s)+\frac{\rho_0\mathbf{u}_0}{L_0}\nabla\cdot(\tilde{\rho}_s\tilde{\phi}_s\tilde{\mathbf{u}}_s)=0,\\
&\frac{\rho_0\mathbf{u}_0}{\tau_0}\frac{\partial}{\partial\tilde{t}}(\tilde{\rho}_s\tilde{\phi}_s\tilde{\mathbf{u}}_s)+\frac{\rho_0\mathbf{u}_0^2}{L_0}\nabla\cdot(\tilde{\rho}_s\tilde{\phi}_s\tilde{\mathbf{u}}_s\tilde{\mathbf{u}}_s)+\frac{p_0}{L_0}\nabla(\tilde{p}_s\tilde{\phi}_s)-\frac{p_0}{L_0}\tilde{p}_I\nabla\tilde{\phi}_s=-\frac{\rho_0\mathbf{u}_0}{L_0}\tilde{\delta}(\tilde{\mathbf{u}}_s-\tilde{\mathbf{u}}_f),\\
&\frac{p_0}{\tau_0}\frac{\partial}{\partial\tilde{t}}(\tilde{\rho}_s\tilde{\phi}_s\tilde{E}_s)+\frac{p_0\mathbf{u}_0}{L_0}\nabla\cdot(\tilde{\phi}_s\tilde{\mathbf{u}}_s(\tilde{\rho}_s\tilde{E}_s+\tilde{p}_s))-\frac{p_0\mathbf{u}_0}{L_0}\tilde{p}_I\tilde{\mathbf{u}}_I\nabla\tilde{\phi}_s=-\frac{p_0^2}{\rho_0\mathbf{u}_0^2\tau_0}\tilde{\mu}\,\tilde{p}_I\,(\tilde{p}_s-\tilde{p}_f)-\frac{\rho_0\mathbf{u}_0^2}{\tau_0}\tilde{\delta}\,\tilde{\mathbf{u}}_I\,(\tilde{\mathbf{u}}_s-\tilde{\mathbf{u}}_f),\\
&\frac{1}{\tau_0}\frac{\partial}{\partial\tilde{t}}\tilde{\phi}_s+\frac{\mathbf{u}_0}{L_0}\tilde{\mathbf{u}}_I\cdot\nabla\tilde{\phi}_s=\frac{p_0}{\rho_0\mathbf{u}_0^2\tau_0}\tilde{\mu}(\tilde{p}_s-\tilde{p}_f).
\end{aligned}\tag{7}$$

Introducing the stiffness parameter $\epsilon=\gamma_0 M^2$, related to a global Mach number $M=|\mathbf{u}_0|/c_0$, with the speed of sound $c_0=\sqrt{\gamma_0 p_0/\rho_0}$, and dropping the tildes for readability, the non-dimensional system finally reads

$$\begin{aligned}
&\frac{\partial}{\partial t}(\rho_f\phi_f)+\nabla\cdot(\rho_f\phi_f\mathbf{u}_f)=0,\\
&\frac{\partial}{\partial t}(\rho_f\phi_f\mathbf{u}_f)+\nabla\cdot(\rho_f\phi_f\mathbf{u}_f\mathbf{u}_f)+\frac{1}{\epsilon}\nabla(p_f\phi_f)+\frac{1}{\epsilon}p_I\nabla\phi_f=\delta(\mathbf{u}_s-\mathbf{u}_f),\\
&\frac{\partial}{\partial t}(\rho_f\phi_f E_f)+\nabla\cdot(\phi_f\mathbf{u}_f(\rho_f E_f+p_f))+p_I\mathbf{u}_I\nabla\phi_f=\frac{1}{\epsilon}\mu\,p_I\,(p_s-p_f)+\epsilon\delta\,\mathbf{u}_I\,(\mathbf{u}_s-\mathbf{u}_f),\\
&\frac{\partial}{\partial t}(\rho_s\phi_s)+\nabla\cdot(\rho_s\phi_s\mathbf{u}_s)=0,\\
&\frac{\partial}{\partial t}(\rho_s\phi_s\mathbf{u}_s)+\nabla\cdot(\rho_s\phi_s\mathbf{u}_s\mathbf{u}_s)+\frac{1}{\epsilon}\nabla(p_s\phi_s)-\frac{1}{\epsilon}p_I\nabla\phi_s=-\delta(\mathbf{u}_s-\mathbf{u}_f),\\
&\frac{\partial}{\partial t}(\rho_s\phi_s E_s)+\nabla\cdot(\phi_s\mathbf{u}_s(\rho_s E_s+p_s))-p_I\mathbf{u}_I\nabla\phi_s=-\frac{1}{\epsilon}\mu\,p_I\,(p_s-p_f)-\epsilon\delta\,\mathbf{u}_I\,(\mathbf{u}_s-\mathbf{u}_f),\\
&\frac{\partial}{\partial t}\phi_s+\mathbf{u}_I\cdot\nabla\phi_s=\frac{1}{\epsilon}\mu(p_s-p_f).
\end{aligned}\tag{8}$$

The above formulation (8) shows that the stiffness is present in the momentum equations, due to the acoustic waves, but also in the energy equations. More precisely, following [43], the energy flux can be split into two contributions, that is

$$\phi_\alpha\mathbf{u}_\alpha(\rho_\alpha E_\alpha+p_\alpha)=\epsilon\rho_\alpha\phi_\alpha k_\alpha\mathbf{u}_\alpha+\rho_\alpha\phi_\alpha h_\alpha\mathbf{u}_\alpha, \tag{9}$$

with the enthalpy defined as

$$h_\alpha = e_\alpha + \frac{p_\alpha}{\rho_\alpha}. \tag{10}$$

Flows at low Mach and high Mach number behave quite differently. For low Mach, the regime is quasi-incompressible, with small pressure variations, and with convection and diffusion domination. On the contrary, for high Mach numbers, the pressure varies significantly and the flow is highly compressible, with the appearance of shock waves. Those behaviors are visible also from the scaled system (8), where the stiffness parameter may assume values $\epsilon \to 0$ or $\epsilon \gg 1$. Along the lines of Re and Abgrall [46], a Mach number common to both phases is identified to expresses the overall compressibility of the mixture flow field. The divergence-free constraint of the velocity field in the low Mach regime will then be recovered for the mixture velocity, see Section 3.6.

3. Numerical scheme

We design a semi-implicit numerical scheme for the solution of the Baer–Nunziato model (1), which is second order accurate in space and time. The method is proven to be well-balanced and asymptotic preserving in the low Mach number limit of the mixture. Firstly, we solve the homogeneous model over a time step Δt, by neglecting the source terms. Next, we apply a relaxation operator, to solve the system of ordinary differential equations (ODE) that is left. Regardless the Mach regime, the time step is unconstrained from the acoustic waves, even in the limit $\epsilon \to 0$, and the scheme obeys the following stability condition:

$$\Delta t \leq \text{CFL} \min_{\Omega} \frac{\varrho}{\max(\max(|\mathbf{u}_f \cdot \mathbf{n}|,\ |\mathbf{u}_s \cdot \mathbf{n}|))}, \tag{11}$$

which is only based on the maximum material speed of the phases.

3.1. First order semi-discrete scheme in time

Let us assume $\mathbf{S}(\mathbf{Q}) = \mathbf{0}$ in (5), and let us adopt the general subscript α for the two phases. The time coordinate is defined in the interval $t \in [0, t_f]$, with t_f the final time. The solution advances in time as $t^{n+1} = t^n + \Delta t$, in which the time step Δt is defined at each iteration according to the CFL condition (11). The discretization in time is based on a semi-implicit formulation, that yields the following scheme:

$$(\rho_\alpha \phi_\alpha)^{n+1} = (\rho_\alpha \phi_\alpha)^n - \Delta t\, \nabla \cdot ((\rho_\alpha \mathbf{u}_\alpha)^n\, \phi_\alpha^{n+1}), \tag{12a}$$

$$(\rho_\alpha \phi_\alpha \mathbf{u}_\alpha)^{n+1} = (\rho_\alpha \phi_\alpha \mathbf{u}_\alpha)^n - \Delta t\, \nabla \cdot ((\rho_\alpha \mathbf{u}_\alpha \mathbf{u}_\alpha)^n\, \phi_\alpha^{n+1}) - \Delta t\, \nabla (\phi_\alpha p_\alpha)^{n+1} + \Delta t\, p_I^n \nabla \phi_\alpha^{n+1}, \tag{12b}$$

$$(\rho_\alpha \phi_\alpha E_\alpha)^{n+1} = (\rho_\alpha \phi_\alpha E_\alpha)^n - \Delta t\, \nabla \cdot ((\rho_\alpha k_\alpha \mathbf{u}_\alpha)^n\, \phi_\alpha^{n+1}) - \Delta t\, \nabla \cdot (h_\alpha^n\, (\rho_\alpha \phi_\alpha \mathbf{u}_\alpha)^{n+1}) + \Delta t\, p_I^n \mathbf{u}_I^n \nabla \phi_\alpha^{n+1}, \tag{12c}$$

$$\phi_s^{n+1} = \phi_s^n - \Delta t\, \mathbf{u}_I^n \nabla \phi_s^{n+1}, \tag{12d}$$

where the splitting of the energy flux (9) in (12c) allows us to treat the part related to the kinetic energy explicitly, while the part related to the enthalpy implicitly. It is evident that the nonlinear convective terms are treated explicitly, and the implicit terms are the stiff ones, involving pressure and volume fraction gradients. However, the volume fraction in the convective fluxes is discretized implicitly in order to ensure the well-balanced property (see Section 3.7). Since the volume fraction Eq. (12d) can be readily solved, the quantity ϕ_s^{n+1} is computed first, and then it can be used in the other equations, making the nonlinear convective fluxes explicitly computable. We underline that the scheme is linearly implicit, meaning that any non-linearity in time is avoided. Indeed, in our semi-implicit scheme the interface quantities are considered explicitly, as well as the enthalpy in the energy flux.

Let us introduce the following abbreviations to account for the explicit contributions:

$$\rho_\alpha^\star = (\rho_\alpha \phi_\alpha)^n - \Delta t\, \nabla \cdot ((\rho_\alpha \mathbf{u}_\alpha)^n\, \phi_\alpha^{n+1}), \tag{13a}$$

$$\mathbf{q}_\alpha^\star = (\rho_\alpha \phi_\alpha \mathbf{u}_\alpha)^n - \Delta t\, \nabla \cdot ((\rho_\alpha \mathbf{u}_\alpha \mathbf{u}_\alpha)^n\, \phi_\alpha^{n+1}), \tag{13b}$$

$$E_\alpha^\star = (\rho_\alpha \phi_\alpha E_\alpha)^n - \Delta t\, \nabla \cdot ((\rho_\alpha k_\alpha \mathbf{u}_\alpha)^n\, \phi_\alpha^{n+1}). \tag{13c}$$

Letting aside the volume fraction Eq. (12d), which is the first one to be solved, and using the above definitions, the remaining system can now be written as

$$(\rho_\alpha \phi_\alpha)^{n+1} = \rho_\alpha^\star, \tag{14a}$$

$$(\rho_\alpha \phi_\alpha \mathbf{u}_\alpha)^{n+1} = \mathbf{q}_\alpha^\star - \Delta t\, \nabla (\phi_\alpha p_\alpha)^{n+1} + \Delta t\, p_I^n \nabla \phi_\alpha^{n+1}, \tag{14b}$$

$$(\rho_\alpha \phi_\alpha e_\alpha)^{n+1} + (\rho_\alpha \phi_\alpha k_\alpha)^{n+1} = E_\alpha^\star - \Delta t\, \nabla \cdot (h_\alpha^n\, (\rho_\alpha \phi_\alpha \mathbf{u}_\alpha)^{n+1}) + \Delta t\, p_I^n \mathbf{u}_I^n \nabla \phi_\alpha^{n+1}. \tag{14c}$$

In the energy Eq. (14c), the total energy at the new time level is split into the internal and the kinetic energy contributions. In particular, to guarantee a linearly implicit scheme, we resort to the semi-implicit linearization proposed in Boscheri and Pareschi [43], hence taking

$$(\rho_\alpha \phi_\alpha k_\alpha)^{n+1} := \frac{1}{2} \frac{(\rho_\alpha \phi_\alpha \mathbf{u}_\alpha)^n}{(\rho_\alpha \phi_\alpha)^n} (\rho_\alpha \phi_\alpha \mathbf{u}_\alpha)^{n+1}. \tag{15}$$

Inserting the momentum Eq. (14b) into the energy Eq. (14c) and expressing the internal energy as a function of the pressure by means of the EOS (2), we obtain a wave equation of the form

$$\begin{aligned} &\frac{(\phi_\alpha p_\alpha)^{n+1}}{\gamma_\alpha - 1} + \frac{\phi_\alpha^{n+1}\gamma_\alpha \pi_\alpha}{\gamma_\alpha - 1} + \frac{1}{2}\frac{(\rho_\alpha\phi_\alpha \mathbf{u}_\alpha)^n}{(\rho_\alpha\phi_\alpha)^n}\left(\mathbf{q}_\alpha^\star - \Delta t \nabla(\phi_\alpha p_\alpha)^{n+1} + \Delta t\, p_I^n \nabla \phi_\alpha^{n+1}\right) = \\ &= E_\alpha^\star - \Delta t\, \nabla\cdot(h_\alpha^n \mathbf{q}_\alpha^\star) + \Delta t^2\, \nabla\cdot(h_\alpha^n \nabla(\phi_\alpha p_\alpha)^{n+1}) - \Delta t^2\, \nabla\cdot(h_\alpha^n p_I^n \nabla\phi_\alpha^{n+1}) + \Delta t\, p_I^n \mathbf{u}_I^n \nabla \phi_\alpha^{n+1}. \end{aligned} \tag{16}$$

Shifting all the unknown pressure elements on the left-hand-side, we get

$$\begin{aligned} &\frac{(\phi_\alpha p_\alpha)^{n+1}}{\gamma_\alpha - 1} + \frac{\phi_\alpha^{n+1}\gamma_\alpha \pi_\alpha}{\gamma_\alpha - 1} - \frac{\Delta t}{2}\frac{(\rho_\alpha\phi_\alpha \mathbf{u}_\alpha)^n}{(\rho_\alpha\phi_\alpha)^n}\nabla(\phi_\alpha p_\alpha)^{n+1} - \Delta t^2\, \nabla\cdot(h_\alpha^n \nabla(\phi_\alpha p_\alpha)^{n+1}) = \\ &= E_\alpha^\star - \frac{1}{2}\mathbf{q}_\alpha^\star \frac{(\rho_\alpha\phi_\alpha \mathbf{u}_\alpha)^n}{(\rho_\alpha\phi_\alpha)^n} - \frac{\Delta t}{2} p_I^n \nabla\phi_\alpha^{n+1}\frac{(\rho_\alpha\phi_\alpha \mathbf{u}_\alpha)^n}{(\rho_\alpha\phi_\alpha)^n} - \Delta t\, \nabla\cdot(h_\alpha^n \mathbf{q}_\alpha^\star) - \Delta t^2\, \nabla\cdot(h_\alpha^n p_I^n \nabla\phi_\alpha^{n+1}) + \Delta t\, p_I^n \mathbf{u}_I^n \nabla\phi_\alpha^{n+1}. \end{aligned} \tag{17}$$

The stiffened gas EOS ensures a linear relation between the internal energy and the pressure, hence yielding a system of linear equations that can be directly solved for the unknown quantities $(\phi_\alpha p_\alpha)^{n+1}$. It must be noted that, using a different EOS, this property may not hold any more, but the pressure formulation keeps the non-linearity only along the diagonal of the system matrix [42,43]. In this case, the nested Newton technique proposed in Brugnano and Casulli [58] can be efficiently applied. The solution of the above system is only meant to provide a stable numerical flux to be used in the conservative update of momentum and total energy according to (14b) and (14c), differently from the non-conservative pressure update proposed in Re and Abgrall [46]. In principle, the wave equation could be solved for the total energy as done in Boscarino et al. [45] for single-phase flows, which however would yield a fully nonlinear system for general EOS. Finally, we remark that the enthalpy must be discretized as

$$h_\alpha^n := \frac{\rho_\alpha^n h_\alpha^n}{\rho_\alpha^{n+1}}, \tag{18}$$

in order to achieve the preservation of constant velocity and pressure flows at the discrete level, as previously studied in [43] for single-phase flows. A more detailed analysis concerning the well-balanced property of the scheme is conducted in Section 3.7.

Remark 1 (On the thermodynamic consistency of the total energy)**.** Once the pressure wave Eq. (17) is solved, the total energy is updated with (14c). By doing so, thermodynamic consistency is violated up to the time order of accuracy of the scheme. Indeed, the new total energy of our semi-discrete scheme is equivalent to

$$(\rho_\alpha\phi_\alpha E_\alpha)^{n+1} = \frac{(\phi_\alpha p_\alpha)^{n+1}}{\gamma_\alpha - 1} + \frac{\phi_\alpha^{n+1}\gamma_\alpha \pi_\alpha}{\gamma_\alpha - 1} + \frac{1}{2}\frac{(\rho_\alpha\phi_\alpha \mathbf{u}_\alpha)^n}{(\rho_\alpha\phi_\alpha)^n}(\rho_\alpha\phi_\alpha \mathbf{u}_\alpha)^{n+1},$$

therefore the error is of order $\mathcal{O}(\Delta t^q)$, with q being the order of the method. To obtain exact thermodynamic consistency, one can avoid the semi-implicit linearization of the kinetic energy (15), which implies the solution of a nonlinear system, and apply a fixed point iteration method, see for instance [42,59]. Another option, recently forwarded in Boscheri et al. [60], is the following one: the pressure system is solved, for the first time, with the proposed linearization of the kinetic energy, for obtaining a *pressure flux*. This pressure will readily be used to update the momentum according to (14b). Next, the pressure system is solved for the second time for the *pressure state*, hence employing the kinetic energy at the new time level which is known at this stage. At the price of two linear systems to be solved, thermodynamic consistency is strictly ensured.

3.2. Second order extension in time

Second order accuracy in time is achieved relying on the class of semi-implicit IMEX schemes forwarded in Boscarino et al. [41]. Here, we limit us to briefly recall the procedure, which starts by writing the governing Eq. (5) under the form of an autonomous system, that is

$$\frac{\partial \mathbf{Q}}{\partial t} = \mathcal{H}(\mathbf{Q}_E(t), \mathbf{Q}_I(t)), \qquad \forall t > 0, \quad \mathbf{Q}_0 = \mathbf{Q}(t = t_0). \tag{19}$$

The function $\mathcal{H}(\mathbf{Q}_E(t), \mathbf{Q}_I(t))$ represents the numerical discretization of the spatial operators, and is partitioned in an explicit and an implicit argument, respectively. The first order semi-discrete scheme presented in the previous section fits the general formulation (19), thus an IMEX Runge–Kutta time marching algorithm can be adopted to achieve second order of accuracy in time. Such schemes are, in general, multi-step methods based on s stages, for which the following Butcher tableaux can be defined:

$$\begin{array}{c|c} \hat{\mathrm{c}} & \hat{\mathrm{A}} \\ \hline & \hat{\mathrm{b}}^T \end{array} \qquad \begin{array}{c|c} \mathrm{c} & \mathrm{A} \\ \hline & \mathrm{b}^T \end{array}, \tag{20}$$

with matrices $(\hat{\mathrm{A}}, \mathrm{A}) \in \mathbb{R}^{s\times s}$ and vectors $(\hat{\mathrm{b}}, \mathrm{b}, \hat{\mathrm{c}}, \mathrm{c}) \in \mathbb{R}^s$. The tableau on the left is applied for the explicit scheme, and the one on the right for the implicit scheme. We use the Stiffly Accurate semi-implicit Runge–Kutta method LSDIRK2(2,2,2) [41]:

$$\begin{array}{c|cc} 0 & 0 & 0 \\ \hat{c} & \hat{c} & 0 \\ \hline & 1-\beta & \beta \end{array} \qquad \begin{array}{c|cc} \beta & \beta & 0 \\ 1 & 1-\beta & \beta \\ \hline & 1-\beta & \beta \end{array}, \tag{21}$$

with $\beta = 1 - 1/\sqrt{2}$ and $\hat{c} = 1/(2\beta)$. For further information and details on the semi-implicit IMEX schemes, the reader is referred to [41], while applications to asymptotic preserving low Mach number schemes can be found, for instance, in Avgerinos et al. [37], Boscheri and Pareschi [43], Boscheri and Tavelli [44].

3.3. Fully discrete scheme in space and time

The discrete computational domain is defined, for $d = 3$, by $\Omega(\mathbf{x}) = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [z_{\min}, z_{\max}]$, and it is paved with a Cartesian grid of characteristic dimensions

$$\Delta x = \frac{x_{\max} - x_{\min}}{N_x} \qquad \Delta y = \frac{y_{\max} - y_{\min}}{N_y} \qquad \Delta z = \frac{z_{\max} - z_{\min}}{N_z}, \tag{22}$$

with $\{N_{\mathbf{x}}, N_{\mathbf{y}}, N_{\mathbf{z}}\}$ representing the number of cells in each spatial direction, hence accounting for a total of $N_e = N_x \times N_y \times N_z$ elements. The index triplet (i, j, k) permits to uniquely define a cell $C_{i,j,k}$ of volume $|C_{i,j,k}| = \Delta x\, \Delta y\, \Delta z$ and center $\mathbf{x}_{i,j,k} = (x_i, y_j, z_k)$. The cell faces in the different directions are labeled with indexes $(i \pm 1/2, j, k)$, $(i, j \pm 1/2, k)$, $(i, j, k \pm 1/2)$, have center points at $\mathbf{x}_{i\pm1/2,j,k} = (\frac{x_i + x_{i\pm1}}{2}, y_j, z_k)$, $\mathbf{x}_{i,j\pm1/2,k} = (x_i, \frac{y_j + y_{j\pm1}}{2}, z_k)$, and $\mathbf{x}_{i,j,k\pm1/2} = (x_i, y_j, \frac{z_k + z_{k\pm1}}{2})$, and normal vectors $\mathbf{n}_x = (1, 0, 0)$, $\mathbf{n}_y = (0, 1, 0)$, and $\mathbf{n}_z = (0, 0, 1)$.

For sake of clarity, we present the numerical scheme for the one-dimensional case considering the x-direction only, bearing in mind that the extension to the remaining spatial directions is straightforward.

The volume fraction Eq. (12d) is the first to be solved, and contains only non-conservative terms. The chosen discretization, in fluctuation form, writes

$$\phi_{s,i}^{n+1} + \frac{\Delta t}{2\Delta x}\left(u_{I,i+1/2}^{n}\,(\phi_{s,i+1}^{n+1} - \phi_{s,i}^{n+1}) - u_{I,i-1/2}^{n}\,(\phi_{s,i-1}^{n+1} - \phi_{s,i}^{n+1})\right) = \phi_{s,i}^{n}, \tag{23}$$

where the interface velocities are evaluated as

$$u_{I,i+1/2}^{n} = \frac{1}{2}\left(u_{I,i+1}^{n} + u_{I,i}^{n}\right), \qquad u_{I,i-1/2}^{n} = \frac{1}{2}\left(u_{I,i}^{n} + u_{I,i-1}^{n}\right), \tag{24}$$

using one of the definitions (3a) and (3b). The above discretization (23) is conceived such that the scheme can preserve a constant pressure and velocity field for a traveling contact wave [46], thus it follows from physical observations. However, it also worth noticing that the same discretization can be retrieved by means of the path-conservative approach (see Parés [61] and references therein) using a simple straight line segment path and a midpoint integration rule for the evaluation of the integral of the jump terms across the element boundaries. The linear system (23) is solved at the aid of the GMRES method [62], where we prescribe a tolerance of 10^{-12} to stop the iterative procedure.

Once the new volume fraction $\phi_{s,i}^{n+1}$ is known, the explicit terms are solved relying on a finite volume method:

$$m_i^{\star} = m_i^n - \frac{\Delta t}{\Delta x}\left(f_{i+1/2}^{m} - f_{i-1/2}^{m}\right), \tag{25}$$

where m represents any variable which we solve for in the Baer–Nunziato equations, and $f_{i\pm1/2}^{m}$ are the numerical fluxes given by

$$\begin{aligned} f_{i+1/2}^{m} &= \frac{1}{2}(f(m_{i+1}^n) + f(m_i^n)) - \frac{1}{2}a_{\Omega}^{n}(m_{i+1}^n - m_i^n), \\ f_{i-1/2}^{m} &= \frac{1}{2}(f(m_i^n) + f(m_{i-1}^n)) - \frac{1}{2}a_{\Omega}^{n}(m_i^n - m_{i-1}^n), \end{aligned} \tag{26}$$

with $f(\cdot)$ being the physical flux related to the m variable. We use a global Lax-Friedrichs approximate Riemann solver, thus the numerical dissipation is computed as

$$a_{\Omega}^{n} = \max_{\Omega}\left(\max(|u_f^n|,\, |u_s^n|)\right), \tag{27}$$

considering only the eigenvalues associated to the convective terms. We remark that the numerical dissipation in the energy equations takes into account the jump of the kinetic energy only, and not the jump of the total energy, which would also contain the internal energy contribution. The finite volume scheme (26) is, thus, used for the computation of the $m^{\star}$ quantities in (13a)–(13c):

$$\rho_{\alpha,i}^{\star} = (\rho_\alpha\phi_\alpha)_i^n - \frac{\Delta t}{\Delta x}\left(f_{i+1/2}^{\rho_\alpha\phi_\alpha} - f_{i-1/2}^{\rho_\alpha\phi_\alpha}\right) = (\rho_\alpha\phi_\alpha)_i^{n+1}, \tag{28a}$$

$$q_{\alpha,i}^{\star} = (\rho_\alpha\phi_\alpha u_\alpha)_i^n - \frac{\Delta t}{\Delta x}\left(f_{i+1/2}^{\rho_\alpha\phi_\alpha u_\alpha} - f_{i-1/2}^{\rho_\alpha\phi_\alpha u_\alpha}\right), \tag{28b}$$

$$E_{\alpha,i}^{\star} = (\rho_\alpha\phi_\alpha E_\alpha)_i^n - \frac{\Delta t}{\Delta x}\left(f_{i+1/2}^{\rho_\alpha\phi_\alpha k_\alpha} - f_{i-1/2}^{\rho_\alpha\phi_\alpha k_\alpha}\right). \tag{28c}$$

For the implicit terms in the momentum and energy Eqs. (14b) and (14c), central finite difference operators are used, which provide second order of accuracy on Cartesian grids. The wave equation for the pressure (17) needs to be solved first, and it is discretized as follows:

$$\begin{aligned}
&\frac{(\phi_\alpha p_\alpha)^{n+1}}{\gamma_\alpha - 1} + \frac{\phi_\alpha^{n+1}\gamma_\alpha \pi_\alpha}{\gamma_\alpha - 1} - \frac{\Delta t}{4\Delta x}\frac{(\rho_\alpha\phi_\alpha u_\alpha)_i^n}{(\rho_\alpha\phi_\alpha)_i^n}\left((\phi_\alpha p_\alpha)_{i+1}^{n+1} - (\phi_\alpha p_\alpha)_{i-1}^{n+1}\right) - \Delta t^2\,\nabla\cdot(h_\alpha^n \nabla(\phi_\alpha p_\alpha)^{n+1}) = \\
&= E_{\alpha,i}^\star - \frac{1}{2}q_{\alpha,i}^\star \frac{(\rho_\alpha\phi_\alpha u_\alpha)_i^n}{(\rho_\alpha\phi_\alpha)_i^n} - \frac{\Delta t}{4\Delta x}\left(p_{I,i+1/2}^n(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i}^{n+1}) - p_{I,i-1/2}^n(\phi_{\alpha,i-1}^{n+1} - \phi_{\alpha,i}^{n+1})\right)\frac{(\rho_\alpha\phi_\alpha u_\alpha)_i^n}{(\rho_\alpha\phi_\alpha)_i^n} \\
&\quad - \frac{\Delta t}{2\Delta x}\left((h_\alpha^n q_\alpha^\star)_{i+1} - (h_\alpha^n q_\alpha^\star)_{i-1}\right) - \Delta t^2\,\nabla\cdot(h_\alpha^n p_I^n \nabla\phi_\alpha^{n+1}) \\
&\quad + \frac{\Delta t}{2\Delta x}\left(p_{I,i+1/2}^n u_{I,i+1/2}^n(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i}^{n+1}) - p_{I,i-1/2}^n u_{I,i-1/2}^n(\phi_{\alpha,i-1}^{n+1} - \phi_{\alpha,i}^{n+1})\right),
\end{aligned} \tag{29}$$

where the same fluctuation form of (23) has been adopted for the non-conservative terms involving the gradient of the volume fraction $\nabla\phi_s^{n+1}$. Let us notice that two terms in the above Eq. (29) are still kept continuous in space, as they need careful attention in their discretization to ensure a compact stencil and the well-balanced property for moving equilibrium solutions. In particular, a consistent discretization of the first term could be

$$\Delta t^2\,\nabla\cdot(h_\alpha^n\nabla(\phi_\alpha p_\alpha)^{n+1}) = \frac{\Delta t^2}{2\Delta x}\left(h_{\alpha,i+1}^n\frac{(\phi_\alpha p_\alpha)_{i+2}^{n+1} - (\phi_\alpha p_\alpha)_i^{n+1}}{2\Delta x} - h_{\alpha,i-1}^n\frac{(\phi_\alpha p_\alpha)_i^{n+1} - (\phi_\alpha p_\alpha)_{i-2}^{n+1}}{2\Delta x}\right). \tag{30}$$

However, this approximation would widen the stencil, as it would span $\{i-2, i-1, i, i+1, i+2\}$. Using Lagrange interpolation polynomials of second degree on the classical stencil $\{i-1, i, i+1\}$, following the approach forwarded in Boscheri and Pareschi [43], would instead lead to the following compact expression:

$$\Delta t^2\,\nabla\cdot(h_\alpha^n\nabla(\phi_\alpha p_\alpha)^{n+1}) = \frac{\Delta t^2}{\Delta x^2}\begin{bmatrix} h_{\alpha,i-1}^n & h_{\alpha,i}^n & h_{\alpha,i+1}^n\end{bmatrix}\begin{bmatrix}3/4 & -1 & 1/4\\ 0 & 0 & 0\\ 1/4 & -1 & 3/4\end{bmatrix}\begin{bmatrix}(\phi_\alpha p_\alpha)_{i-1}^{n+1}\\ (\phi_\alpha p_\alpha)_i^{n+1}\\ (\phi_\alpha p_\alpha)_{i+1}^{n+1}\end{bmatrix} + \mathcal{O}(\Delta x^2), \tag{31}$$

which is still second order accurate. The second term $\Delta t^2\,\nabla\cdot(h_\alpha^n p_I^n\nabla\phi_\alpha^{n+1})$ is approximated by means of the same second order discrete operator (31), which is proven to guarantee constant pressure and velocity preservation across a contact discontinuity, as detailed in Section 3.7.

We solve the pressure system (29) directly for the quantity $(\phi_\alpha p_\alpha)^{n+1}$ using again the GMRES solver. Next, the momentum (14b), and then the energy (14c) equations are finally updated:

$$(\rho_\alpha\phi_\alpha u_\alpha)_i^{n+1} = q_{\alpha,i}^\star - \frac{\Delta t}{2\Delta x}((\phi_\alpha p_\alpha)_{i+1}^{n+1} - (\phi_\alpha p_\alpha)_{i-1}^{n+1}) + \frac{\Delta t}{2\Delta x}\left(p_{I,i+1/2}^n(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i}^{n+1}) - p_{I,i-1/2}^n(\phi_{\alpha,i-1}^{n+1} - \phi_{\alpha,i}^{n+1})\right) \tag{32a}$$

$$\begin{aligned}(\rho_\alpha\phi_\alpha E_\alpha)_i^{n+1} &= E_{\alpha,i}^\star - \frac{\Delta t}{2\Delta x}\left((h_\alpha^n(\rho_\alpha\phi_\alpha\mathbf{u}_\alpha)^{n+1})_{i+1} - (h_\alpha^n(\rho_\alpha\phi_\alpha\mathbf{u}_\alpha)^{n+1})_{i-1}\right) + \\
&+ \frac{\Delta t}{2\Delta x}\left(p_{I,i+1/2}^n u_{I,i+1/2}^n(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i}^{n+1}) - p_{I,i-1/2}^n u_{I,i-1/2}^n(\phi_{\alpha,i-1}^{n+1} - \phi_{\alpha,i}^{n+1})\right).\end{aligned} \tag{32b}$$

3.4. Second order numerical dissipation

As the central finite difference method is inherently second order accurate on a Cartesian grid of uniform spacing Δx, no other action is needed for the spatial operators defined in Section 3.3. However, particular care must be paid to the definition of the numerical viscosity in the finite volume fluxes (26), as that must also be of order $\mathcal{O}(\Delta x^2)$ to preserve the scheme's overall accuracy. Therefore, any generic cell centered quantity m_i^n undergoes a linear reconstruction procedure to compute the polynomial $w_i^n(x)$ defined as

$$w_i^n(x) := m_i^n + (x - x_i)\,\alpha_{x,i}. \tag{33}$$

The unknown reconstruction coefficient $\alpha_{x,i}$ is evaluated using the minmod limiter, that is

$$\alpha_{x,i} = \begin{cases} 0 & \text{if} \quad \alpha_{x,\ell}\,\alpha_{x,err} \le 0 \\ \alpha_{x,\ell} & \text{if} \quad |\alpha_{x,\ell}| \le |\alpha_{x,err}| \\ \alpha_{x,err} & \text{if} \quad |\alpha_{x,\ell}| > |\alpha_{x,err}| \end{cases}, \qquad \alpha_{x,\ell} = \frac{m_i^n - m_{i-1}^n}{\Delta x}, \quad \alpha_{x,err} = \frac{m_{i+1}^n - m_i^n}{\Delta x}. \tag{34}$$

The numerical dissipation jumps in (26) are then given in terms of the second order boundary extrapolated data within each cell, hence

$$D_{i+1/2}^m = \frac{1}{2}a_\Omega^n(w_{i+1}^n(x_{i+1/2}) - w_i^n(x_{i+1/2})), \qquad D_{i-1/2}^m = \frac{1}{2}a_\Omega^n(w_i^n(x_{i-1/2}) - w_{i-1}^n(x_{i-1/2})), \tag{35}$$

while letting the central flux contribution untouched, as written in (26). The same procedure applies whenever an artificial viscosity term is added to the scheme.

Numerical dissipation in the volume fraction equation. For strong gradients in the volume fraction distribution, an additional stabilization term might be needed in (23), that is taken explicitly on the right hand side:

$$\phi_{s,i}^{n+1} + \frac{\Delta t}{2\Delta x}\left(u_{I,i+1/2}^{n}(\phi_{s,i+1}^{n+1}-\phi_{s,i}^{n+1}) - u_{I,i-1/2}^{n}(\phi_{s,i-1}^{n+1}-\phi_{s,i}^{n+1})\right) = \phi_{s,i}^{n} + \frac{\Delta t}{\Delta x^2}\left(\nu^{+}(\phi_{s,i+1}^{n}-\phi_{s,i}^{n}) - \nu^{-}(\phi_{s,i}^{n}-\phi_{s,i-1}^{n})\right), \tag{36}$$

with a viscosity coefficient chosen as

$$\nu^{\pm}(\sigma) = \frac{\sigma_i^n + \sigma_{i\pm1}^n}{2}, \qquad \sigma_i^n = \frac{1}{2}\Delta x\, a_{\Omega}^n. \tag{37}$$

We remark that the artificial viscosity contribution introduced in (36) must be accounted for also in the pressure system (29), and in the conservative update of the total energy (32b), with the following rescaled dissipation coefficients $\nu^{\pm}(\sigma_\star)$:

$$\sigma_{i,\star}^n = \sigma_i^n \frac{\partial(\rho_\alpha\phi_\alpha e_\alpha)_i^n}{\partial\phi_{\alpha,i}^n}. \tag{38}$$

More details can be found in Section 3.7, where the well-balanced property of the scheme is shown.

Numerical dissipation in the energy equation. For further stabilization at high Mach number flows, the convective fluxes (26) related to the energy equation may be supplemented with an artificial viscosity term on the pressure jump:

$$\begin{aligned} f_{i+1/2}^{\rho_\alpha\phi_\alpha k_\alpha} =& \frac{1}{2}\left(((\rho_\alpha k_\alpha u_\alpha)^n \phi_\alpha^{n+1})_{i+1} + ((\rho_\alpha k_\alpha u_\alpha)^n\phi_\alpha^{n+1})_i \right) - \frac{1}{2}a_\Omega^n\left((\rho_\alpha\phi_\alpha k_\alpha)_{i+1}^n - (\rho_\alpha\phi_\alpha k_\alpha)_i^n \right) \\ &- \frac{1}{\Delta x}\,\nu^{+}(\sigma_{\star\star})\left(p_{\alpha,i+1}^n - p_{\alpha,i}^n\right), \end{aligned} \tag{39}$$

with the rescaled coefficients $\nu^{\pm}(\sigma_{\star\star})$ given by

$$\sigma_{i,\star\star}^n = \sigma_i^n \frac{\partial(\rho_\alpha\phi_\alpha e_\alpha)_i^n}{\partial p_{\alpha,i}^n}. \tag{40}$$

3.5. Relaxation solver

Until this point, the numerical scheme has been presented for the homogeneous version of the Baer–Nunziato system, thus the new solution obtained with the semi-implicit scheme detailed in Section 3.3 is labeled with $\mathbf{Q}^H$, according to the compact notation introduced in (5). A Strang splitting scheme [63] is here adopted, thus we aim at solving the remaining ODE system:

$$\frac{\mathbf{Q}_i^{n+1} - \mathbf{Q}_i^H}{\Delta t} = \mathbf{S}(\mathbf{Q}_i^{n+1}), \qquad i = 1, \dots, N_x, \tag{41}$$

where the vector of potentially stiff source terms is discretized using again a semi-implicit scheme, since we want to avoid any non-linearity. Specifically, only in the x-direction, the sources are discretized as

$$\mathbf{S}(\mathbf{Q}_i^{n+1}) = \begin{bmatrix} 0 \\ \delta\,(u_{s,i}^{n+1} - u_{f,i}^{n+1}) \\ \mu\, p_{I,i}^H\,(p_{s,i}^H - p_{f,i}^H) + \delta\, u_{I,i}^H\,(u_{s,i}^{n+1} - u_{f,i}^{n+1}) \\ 0 \\ -\delta\,(u_{s,i}^{n+1} - u_{f,i}^{n+1}) \\ -\mu\, p_{I,i}^H\,(p_{s,i}^H - p_{f,i}^H) - \delta\, u_{I,i}^H\,(u_{s,i}^{n+1} - u_{f,i}^{n+1}) \\ \mu\,(p_{s,i}^H - p_{f,i}^H) \end{bmatrix}. \tag{42}$$

The pressure terms are directly taken from the solution of the wave Eq. (29), thus they are readily computable. Inspired from Re and Abgrall [46], the new velocities $(u_{s,i}^{n+1}, u_{f,i}^{n+1})$ are evaluated by solving a linear algebraic system, locally defined within each computational cell. Once the velocities have been obtained, the sources in the energy equations are finally computed. Second order of accuracy in space is automatically achieved, since the computations are performed at the cell center, which corresponds to the midpoint quadrature rule. For time accuracy, we resort to the semi-implicit IMEX scheme (21).

Summary of the linearly semi-implicit scheme. The numerical scheme is now complete, and it can be summarized by the following steps.

1. Solve implicitly the equation of the volume fraction (23) with the GMRES iterative method, and obtain ϕ_s^{n+1}.
2. Solve explicitly the convective fluxes with the finite volume scheme (25) and (26), and obtain $\rho_\alpha^\star$, $q_\alpha^\star$, and $E_\alpha^\star$. We recall that $\rho_\alpha^\star = (\rho_\alpha\phi_\alpha)^{n+1}$.
3. Solve the linearly implicit wave Eq. (29) directly for $(\phi_\alpha p_\alpha)^{n+1}$ with the GMRES iterative method.
4. Update the momentum equations according to (32a), and obtain $(\rho_\alpha\phi_\alpha u_\alpha)^{n+1}$.
5. Update the energy equations with (32b), and obtain $(\rho_\alpha\phi_\alpha E_\alpha)^{n+1}$.
6. Starting from the previously obtained solution of the homogeneous system, relax the system through the source terms, evaluated with the semi-implicit discretization (42).

3.6. Asymptotic preserving property

Given a mathematical model characterized by a stiffness parameter ϵ, a numerical method which solves this problem over time with step size Δt is considered Asymptotic Preserving (AP) if it retrieves, at the discrete level, a consistent and stable discretization of the limit model obtained for $\epsilon \to 0$, for any Δt independent of ϵ. For single-phase ideal compressible gases, this means that an AP scheme can preserve the correct asymptotic transition from the compressible to the incompressible equations, as the Mach number tends to zero.

The numerical method presented in this work is AP, as it remains accurate and stable in the stiff regime. To prove it, we first derive the asymptotic limit of the continuous system of equations, and then we analyze the temporal semi-discrete scheme to show that it leads to a consistent discretization of the incompressible model for $\epsilon \to 0$.

Limit model of the mixture equations. Apart from the pioneering analysis of Varsakelis and Papalexandris [64], low Mach asymptotics have not yet been generalized to multi-phase flow models. It is yet not clear how to derive asymptotic limits for multi-phase flows, since the Mach number can be defined either separately for each phase, or uniquely for the mixture. Here, we analyze the low Mach number limit of the mixture model. Let us assume a generic bounded domain Ω, with impermeable boundary conditions on $\partial\Omega$

$$\mathbf{u}_\alpha \cdot \mathbf{n} = 0, \tag{43}$$

where $\mathbf{n}$ denotes the unit outward normal vector to the boundary $\partial\Omega$, and let the kth order power expansion for the generic variable m be given in terms of the parameter ϵ as

$$m = m_{(0)} + \epsilon m_{(1)} + \epsilon^2 m_{(2)} + \ldots + \mathcal{O}(\epsilon^k). \tag{44}$$

By assuming well-prepared initial data, all the variables of the non-dimensional governing Eq. (8) can be expanded with respect to the common stiffness parameter ϵ according to (44).

Substituting the expansion (44) in the non-dimensional volume fraction equation of (8) and assuming a finite relaxation parameter δ, one immediately obtains

$$\mathcal{O}(\epsilon^{-1}): \quad \mu(p_{s_{(0)}} - p_{f_{(0)}}) = 0 \quad \Rightarrow \quad p_{s_{(0)}} = p_{f_{(0)}} = p_{I_{(0)}} = p_{(0)}, \tag{45}$$

and considering that ϕ_f is governed by the same equation as ϕ_s, in general we have

$$\mathcal{O}(\epsilon^{0}): \quad \partial_t \phi_{\alpha_{(0)}} + u_{I_{(0)}} \nabla \phi_{\alpha_{(0)}} = \pm\mu(p_{s_{(1)}} - p_{f_{(1)}}). \tag{46}$$

The continuity equation at the 0−th leading order reads

$$\partial_t(\rho_{\alpha_{(0)}} \phi_{\alpha_{(0)}}) + \nabla \cdot (\rho_{\alpha_{(0)}} \phi_{\alpha_{(0)}} \mathbf{u}_{\alpha_{(0)}}) = 0. \tag{47}$$

By collecting like powers of ϵ in the momentum equation, we have

$$\mathcal{O}(\epsilon^{-1}): \quad \nabla(p_{\alpha_{(0)}} \phi_{\alpha_{(0)}}) - p_{I_{(0)}} \nabla \phi_{\alpha_{(0)}} = 0, \tag{48a}$$

$$\mathcal{O}(\epsilon^{\ 0}): \quad \partial_t(\rho_{\alpha_{(0)}} \phi_{\alpha_{(0)}} \mathbf{u}_{\alpha_{(0)}}) + \nabla \cdot (\rho_{\alpha_{(0)}} \phi_{\alpha_{(0)}} \mathbf{u}_{\alpha_{(0)}} \mathbf{u}_{\alpha_{(0)}}) + \nabla(p_{\alpha_{(1)}} \phi_{\alpha_{(1)}}) - p_{I_{(1)}} \nabla \phi_{\alpha_{(1)}} = \mp\delta(\mathbf{u}_{s_{(0)}} - \mathbf{u}_{f_{(0)}}). \tag{48b}$$

Summing Eq. (48a) for both phases, using the result of (45), and recalling that the volume fractions sum up to unity, one gets

$$\begin{aligned}
\nabla(p_{s_{(0)}} \phi_{s_{(0)}} + p_{f_{(0)}} \phi_{f_{(0)}}) &= p_{I_{(0)}} \nabla(\phi_{s_{(0)}} + \phi_{f_{(0)}}), \\
\nabla(p_{(0)}(\phi_{s_{(0)}} + \phi_{f_{(0)}})) &= p_{(0)} \cdot 0, \\
\nabla p_{(0)} &= 0 \quad \Rightarrow \quad p_{(0)} = p_{(0)}(t),
\end{aligned} \tag{49}$$

which shows that, at the 0−th leading order, the pressure is constant in space, and changes in time only due to boundary conditions. Finally, from the energy equation, we get

$$\mathcal{O}(\epsilon^{0}): \quad \partial_t(\rho_{\alpha_{(0)}} \phi_{\alpha_{(0)}} e_{\alpha_{(0)}}) + \nabla \cdot (\rho_{\alpha_{(0)}} \phi_{\alpha_{(0)}} h_{\alpha_{(0)}} \mathbf{u}_{\alpha_{(0)}}) - p_{I_{(0)}} \mathbf{u}_{I_{(0)}} \nabla \phi_{\alpha_{(0)}} = \mp\mu p_{I_{(1)}}(p_{s_{(1)}} - p_{f_{(1)}}), \tag{50a}$$

$$\mathcal{O}(\epsilon^{1}): \quad \partial_t(\rho_{\alpha_{(0)}} \phi_{\alpha_{(0)}} k_{\alpha_{(0)}}) + \nabla \cdot (\rho_{\alpha_{(0)}} \phi_{\alpha_{(0)}} k_{\alpha_{(0)}} \mathbf{u}_{\alpha_{(0)}}) = \mp\delta \mathbf{u}_{I_{(0)}}(\mathbf{u}_{s_{(0)}} - \mathbf{u}_{f_{(0)}}). \tag{50b}$$

For the sake of simplicity, we assume now to deal with the ideal gas EOS, hence setting $\pi_\alpha = 0$ in Eq. (2). Substituting the definitions of internal energy and enthalpy as functions of pressure and density, and integrating Eq. (50a) on the domain Ω, we obtain

$$\frac{1}{\gamma_\alpha - 1} \int_\Omega \frac{\mathrm{d}(\phi_{\alpha_{(0)}} p_{(0)})}{\mathrm{d}t} \, d\mathbf{x} + \frac{\gamma_\alpha}{\gamma_\alpha - 1} p_{(0)} \int_{\partial\Omega} \phi_{\alpha_{(0)}} \mathbf{u}_{\alpha_{(0)}} \cdot \mathbf{n} \, dS - p_{(0)} \int_\Omega \mathbf{u}_{I_{(0)}} \nabla \phi_{\alpha_{(0)}} \, d\mathbf{x} = \mp\mu \int_\Omega p_{I_{(1)}}(p_{s_{(1)}} - p_{f_{(1)}}) \, d\mathbf{x}. \tag{51}$$

Now, summing the above equation for both phases, using the impermeable boundary conditions (43), and inserting (46) to compute the relaxation source terms, lead to

$$\frac{\gamma_f + \gamma_s - 2}{(\gamma_s - 1)(\gamma_f - 1)} |\Omega| \frac{\mathrm{d}p_{(0)}}{\mathrm{d}t} - p_{(0)} \int_\Omega \mathbf{u}_{I_{(0)}} \nabla(\phi_{s_{(0)}} + \phi_{f_{(0)}}) \, d\mathbf{x} = -\int_\Omega p_{I_{(1)}} u_{I_{(0)}} \nabla(\phi_{s_{(0)}} + \phi_{f_{(0)}}) \, d\mathbf{x}, \tag{52}$$

thus all terms vanish except for

$$\frac{\mathrm{d}p_{(0)}}{\mathrm{d}t} = 0, \tag{53}$$

which suggests that, for the mixture, the pressure $p_{(0)}$ is a constant both in space and time. Considering again Eq. (50a) for the mixture, and using the last result on the pressure, we remain with

$$p_{(0)}\nabla\cdot(\phi_{s_0}\mathbf{u}_{s_0}+\phi_{f_0}\mathbf{u}_{f_0})=0. \tag{54}$$

The above equation shows that the mixture velocity is divergence-free, which is typical of an incompressible flow. Moreover, taking into account that the relaxation processes are instantaneous, or at least faster than the scale associated to the homogeneous part of the system, we can say that $\mathbf{u}_{s_0}=\mathbf{u}_{f_0}=\mathbf{u}_{I_0}=\mathbf{u}_{(0)}$. With this hypothesis, Eq. (54) takes the classical form

$$\nabla\cdot\mathbf{u}_{(0)}=0. \tag{55}$$

We underline that the resulting system composed by this last equation, which is indeed the energy equation, and the leading order continuity (47) and momentum (48b) equations computed for the mixture, represents the incompressible Euler model, in the case of a single-phase flow. This is the asymptotic limit of the continuous model for the mixture.

Asymptotic preserving scheme of the limit mixture model. We consider the semi-discrete first order scheme (12a)–(12d), bearing in mind that the second order IMEX method preserves the asymptotic preserving property and is asymptotically accurate [41]. Mimicking the analysis carried out at the continuous level, we start by inserting the expansion (44) in the semi-discrete scheme, with the addition of the source terms, discretized according to (42). Well-prepared initial data are thus assumed at the discrete level as well. From the volume fraction Eq. (12d), we obtain

$$\mathcal{O}(\epsilon^{-1}):\quad \mu(p^H_{s_{(0)}}-p^H_{f_{(0)}})=0 \quad\Rightarrow\quad p^H_{s_{(0)}}=p^H_{f_{(0)}}=p^H_{I_{(0)}}=p^H_{(0)}. \tag{56}$$

In the same way, from the momentum Eq. (12b), it follows that the pressure is constant in space, meaning that Eq. (49) holds true also at the semi-discrete level:

$$\begin{aligned}&\nabla(p^H_{s_{(0)}}\phi^{n+1}_{s_{(0)}}+p^H_{f_{(0)}}\phi^{n+1}_{f_{(0)}})=p^n_{I_{(0)}}\nabla(\phi_{s_{(0)}}+\phi_{f_{(0)}})^{n+1},\\ &\quad\nabla(p^H_{(0)}(\phi^{n+1}_{s_{(0)}}+\phi^{n+1}_{f_{(0)}}))=p^n_{I_{(0)}}\cdot 0,\quad\Rightarrow\quad \nabla p^H_{(0)}=0.\end{aligned} \tag{57}$$

Therefore, the entire system, at the 0−th leading order, is discretized as follows:

$$(\rho_{\alpha_{(0)}}\phi_{\alpha_{(0)}})^{n+1}=(\rho_{\alpha_{(0)}}\phi_{\alpha_{(0)}})^n-\Delta t\,\nabla\cdot((\rho_{\alpha_{(0)}}\mathbf{u}_{\alpha_{(0)}})^n\,\phi^{n+1}_{\alpha_{(0)}}), \tag{58a}$$

$$(\rho_{\alpha_{(0)}}\phi_{\alpha_{(0)}}\mathbf{u}_{\alpha_{(0)}})^{n+1}=(\rho_{\alpha_{(0)}}\phi_{\alpha_{(0)}}\mathbf{u}_{\alpha_{(0)}})^n-\Delta t\,\nabla\cdot((\rho_{\alpha_{(0)}}\mathbf{u}_{\alpha_{(0)}}\mathbf{u}_{\alpha_{(0)}})^n\,\phi^{n+1}_{\alpha_{(0)}})-\Delta t\,\nabla(\phi^{n+1}_{\alpha_{(1)}}p^H_{\alpha_{(1)}})+\Delta t\,p^n_{I_{(1)}}\nabla\phi^{n+1}_{\alpha_{(1)}}\mp\Delta t\,\delta(\mathbf{u}^{n+1}_{s_{(0)}}-\mathbf{u}^{n+1}_{f_{(0)}}), \tag{58b}$$

$$(\rho_{\alpha_{(0)}}\phi_{\alpha_{(0)}}e_{\alpha_{(0)}})^{n+1}=(\rho_{\alpha_{(0)}}\phi_{\alpha_{(0)}}e_{\alpha_{(0)}})^n-\Delta t\,\nabla\cdot(h^n_{\alpha_{(0)}}(\rho_{\alpha_{(0)}}\phi_{\alpha_{(0)}}\mathbf{u}_{\alpha_{(0)}})^{n+1})+\Delta t\,p^n_{I_{(0)}}\mathbf{u}^n_{I_{(0)}}\nabla\phi^{n+1}_{\alpha_{(0)}}\mp\Delta t\,\mu p^{n+1}_{I_{(1)}}(p^H_{s_{(1)}}-p^H_{f_{(1)}}), \tag{58c}$$

$$\phi^{n+1}_{\alpha_{(0)}}=\phi^n_{\alpha_{(0)}}-\Delta t\,\mathbf{u}^n_{I_{(0)}}\nabla\phi^{n+1}_{\alpha_{(0)}}\pm\Delta t\,\mu(p^H_{s_{(1)}}-p^H_{f_{(1)}}), \tag{58d}$$

while the kinetic terms drop from the energy equation, since they are of first order in the expansion:

$$\mathcal{O}(\epsilon^1):\quad (\rho_{\alpha_{(0)}}\phi_{\alpha_{(0)}}k_{\alpha_{(0)}})^{n+1}=(\rho_{\alpha_{(0)}}\phi_{\alpha_{(0)}}k_{\alpha_{(0)}})^n-\Delta t\,\nabla\cdot((\rho_{\alpha_{(0)}}k_{\alpha_{(0)}}\mathbf{u}_{\alpha_{(0)}})^n\,\phi^{n+1}_{\alpha_{(0)}})\mp\delta\mathbf{u}^{n+1}_{I_{(0)}}(\mathbf{u}^{n+1}_{s_{(0)}}-\mathbf{u}^{n+1}_{f_{(0)}}). \tag{59}$$

Using the definitions of internal energy and enthalpy as functions of pressure and density with the ideal gas EOS ($\pi_\alpha=0$ in Eq. (2)), in addition to the discretization of the enthalpy according to (18), the energy Eq. (58c) is integrated on the computational domain, hence yielding

$$p^{n+1}_{(0)}\int_\Omega\phi^{n+1}_{\alpha_0}\,d\mathbf{x}=p^n_{(0)}\int_\Omega\phi^n_{\alpha_0}\,d\mathbf{x}-\Delta t p^n_{(0)}+\int_{\partial\Omega}\phi^{n+1}_{\alpha_0}\mathbf{u}^{n+1}_{\alpha_0}\cdot\mathbf{n}\,dS+\Delta t\,p^n_{(0)}\int_\Omega\mathbf{u}^n_{I_0}\nabla\phi^{n+1}_{\alpha_0}\,d\mathbf{x}\mp\Delta t\,\mu\int_\Omega p^H_{I_1}(p^H_{s_1}-p^H_{f_1})\,d\mathbf{x}. \tag{60}$$

For the mixture, we sum the equation for both phases, apply the impermeable boundary conditions, and use the relation (56), so that we are left with the discrete analogous of Eq. (53):

$$p^{n+1}_{(0)}=p^n_{(0)}, \tag{61}$$

which means that the pressure at the leading order remains constant for all the time steps. Using the last result to rewrite the energy equation, we have

$$p_{(0)}\phi^{n+1}_{\alpha_{(0)}}=p_{(0)}\phi^n_{\alpha_{(0)}}-\Delta t p_{(0)}\,\nabla\cdot(\phi_{\alpha_{(0)}}\mathbf{u}_{\alpha_{(0)}})^{n+1}+\Delta t\,p_{(0)}\mathbf{u}^n_{I_{(0)}}\nabla\phi^{n+1}_{\alpha_{(0)}}\mp\Delta t\,\mu p^H_{I_1}(p^H_{s_{(1)}}-p^H_{f_{(1)}}), \tag{62}$$

which, for the mixture, yields $\nabla\cdot(\phi_{s_{(0)}}\mathbf{u}_{s_{(0)}}+\phi_{f_{(0)}}\mathbf{u}_{f_{(0)}})^{n+1}=0$. Therefore, we obtain the semi-discrete version of (54), and the scheme is asymptotic preserving.

Remark 2. When the phases are treated independently rather than as a mixture, the relaxation terms vanish. However, the results presented here cannot be obtained under such conditions. In that case, a different approach would be required, involving the definition of distinct Mach numbers for each phase, and treating them separately [51]. For the single-phase case, the above discrete asymptotic analysis exactly retrieves the semi-discrete AP scheme proposed in Boscheri and Pareschi [43] for the Euler equations of compressible gas dynamics.

The results of this section prove that the numerical scheme preserves the asymptotic behavior of the mixture problem at the discrete level. However, it is important to assure that the numerical dissipation does not dominate for small values of ϵ, as studied in Dellacherie [22]. As discussed in Section 3.4, the numerical viscosity is only proportional to the material velocity of the phases, so it is independent of ϵ by construction. The additional dissipation (39) given in terms of the pressure jumps is only activated for very high Mach number flows.

3.7. Well-balanced discretization property

A numerical scheme is well-balanced if it preserves some steady-state solutions of the governing equations also at the discrete level. Hereafter, we show that the proposed semi-implicit scheme for the Baer–Nunziato model is well-balanced, as it maintains a constant pressure and velocity field across a moving contact discontinuity [65]. This means that a multi-phase flow, traveling at constant velocity and pressure for all phases, must only transport the volume fraction density, with no other perturbations.

The following proof is treated for the one-dimensional case, but it easily extends to the three-dimensional case. Let $\Omega = [x_L, x_R]$ be the computational domain, with $x_D \in \Omega$ being the position of the initial discontinuity. The initial condition is defined at $t = t_0$, for all cells $i = 1, \dots, N_x$, and it writes

$$\rho_{\alpha,i}(t_0) = \begin{cases} \rho_{\alpha_L} & x \leq x_D \\ \rho_{\alpha_R} & x > x_D \end{cases}, \quad \phi_{\alpha,i}(t_0) = \begin{cases} \phi_{\alpha_L} & x \leq x_D \\ \phi_{\alpha_R} & x > x_D \end{cases}, \quad u_{\alpha,i}(t_0) = u_0, \quad p_{\alpha,i}(t_0) = p_0, \tag{63}$$

with p_0 and u_0 being a constant pressure and velocity field, respectively. Moreover, $\rho_{\alpha_L} \neq \rho_{\alpha_R}$ and $\phi_{\alpha_L} \neq \phi_{\alpha_R}$ are non-negative real numbers. This initial condition corresponds to a stationary moving contact wave, since the source terms cancel out, leaving the homogeneous system of equations only. The solution of the convective sub-system (28) leads to

$$\begin{aligned} \rho^\star_{\alpha,i} &= (\rho_\alpha \phi_\alpha)^n_i - \frac{\Delta t}{\Delta x}\left(f^{\rho_\alpha\phi_\alpha}_{i+1/2} - f^{\rho_\alpha\phi_\alpha}_{i-1/2} \right)^n \\ &:= (\rho_\alpha\phi_\alpha)^{n+1}_i, \end{aligned} \tag{64a}$$

$$q^\star_{\alpha,i} = (\rho_\alpha\phi_\alpha u_\alpha)^n_i - \frac{\Delta t}{\Delta x}\left(f^{\rho_\alpha\phi_\alpha u_\alpha}_{i+1/2} - f^{\rho_\alpha\phi_\alpha u_\alpha}_{i-1/2} \right)^n = u_0\left((\rho_\alpha\phi_\alpha)^n_i - \frac{\Delta t}{\Delta x}\left(f^{\rho_\alpha\phi_\alpha}_{i+1/2} - f^{\rho_\alpha\phi_\alpha}_{i-1/2} \right)^n \right) = u_0 (\rho_\alpha\phi_\alpha)^{n+1}_i, \tag{64b}$$

$$\begin{aligned} E^\star_{\alpha,i} &= (\rho_\alpha\phi_\alpha e_\alpha)^n_i + \frac{u_0^2}{2}(\rho_\alpha\phi_\alpha)^n_i - \frac{\Delta t}{\Delta x}\left(f^{\rho_\alpha\phi_\alpha k_\alpha}_{i+1/2} - f^{\rho_\alpha\phi_\alpha k_\alpha}_{i-1/2} \right)^n \\ &= (\rho_\alpha\phi_\alpha e_\alpha)^n_i + \frac{u_0^2}{2}(\rho_\alpha\phi_\alpha)^n_i - \frac{u_0^2}{2}\frac{\Delta t}{\Delta x}\left(f^{\rho_\alpha\phi_\alpha}_{i+1/2} - f^{\rho_\alpha\phi_\alpha}_{i-1/2} \right)^n = (\rho_\alpha\phi_\alpha e_\alpha)^n_i + \frac{u_0^2}{2}(\rho_\alpha\phi_\alpha)^{n+1}_i, \end{aligned} \tag{64c}$$

$$\begin{aligned} \phi^\star_{\alpha,i} &= \phi^n_{\alpha,i} - \frac{\Delta t}{2\Delta x} u_0 \left(\phi^{n+1}_{\alpha,i+1} - \phi^{n+1}_{\alpha,i-1} \right) \\ &:= \phi^{n+1}_{\alpha,i}. \end{aligned} \tag{64d}$$

So far, no discretization of the non-conservative terms has been carried out, as it would not be possible to exactly balance the second order terms in the pressure sub-system by simple substitution. Recalling the wave equation for the pressure (17) and restricting ourselves to the ideal gas EOS with $\pi_\alpha = 0$ in (2), one gets

$$\begin{aligned} & p_0 \frac{\phi^{n+1}_{\alpha,i}}{\gamma - 1} - \underline{\frac{\Delta t}{4\Delta x} u_0 p_0 \left(\phi^{n+1}_{\alpha,i+1} - \phi^{n+1}_{\alpha,i-1} \right)} \\ & \underline{- \frac{\Delta t^2}{\Delta x^2} p_0 \left(\left(\frac{3}{4} h^n_{\alpha,i-1} + \frac{1}{4} h^n_{\alpha,i+1} \right) \phi^{n+1}_{\alpha,i-1} - \left(h^n_{\alpha,i-1} + h^n_{\alpha,i+1} \right) \phi^{n+1}_{\alpha,i} + \left(\frac{1}{4} h^n_{\alpha,i-1} + \frac{3}{4} h^n_{\alpha,i+1} \right) \phi^{n+1}_{\alpha,i+1} \right)} \\ & = p_0 \frac{\phi^n_{\alpha,i}}{\gamma - 1} + \underline{\frac{u_0^2}{2}(\rho_\alpha\phi_\alpha)^{n+1}_i} - \underline{\frac{u_0^2}{2}(\rho_\alpha\phi_\alpha)^{n+1}_i} - \underline{\frac{\Delta t}{4\Delta x} p_0 u_0 \left(\phi^{n+1}_{\alpha,i+1} - \phi^{n+1}_{\alpha,i-1} \right)} \\ & \quad - \frac{\Delta t}{2\Delta x} u_0 \left((\rho_\alpha\phi_\alpha)^{n+1}_{i+1} h^n_{\alpha,i+1} - (\rho_\alpha\phi_\alpha)^{n+1}_{i-1} h^n_{\alpha,i-1} \right) + \frac{\Delta t}{2\Delta x} p_0 u_0 \left(\phi^{n+1}_{\alpha,i+1} - \phi^{n+1}_{\alpha,i-1} \right) \\ & \quad \underline{- \frac{\Delta t^2}{\Delta x^2} p_0 \left(\left(\frac{3}{4} h^n_{\alpha,i-1} + \frac{1}{4} h^n_{\alpha,i+1} \right) \phi^{n+1}_{\alpha,i-1} - \left(h^n_{\alpha,i-1} + h^n_{\alpha,i+1} \right) \phi^{n+1}_{\alpha,i} + \left(\frac{1}{4} h^n_{\alpha,i-1} + \frac{3}{4} h^n_{\alpha,i+1} \right) \phi^{n+1}_{\alpha,i+1} \right)}. \end{aligned} \tag{65}$$

As previously mentioned, the term $\nabla \cdot \left(h^n_\alpha p^n_I \nabla \phi^{n+1}_\alpha \right)$ on the right-hand-side of the above equation has been discretized in the same manner as the term $\nabla \cdot \left(h^n_\alpha \nabla (\phi_\alpha p_\alpha)^{n+1} \right)$ on the left-hand-side, thus resorting to the compact finite difference operator (31). This choice allows the two underlined terms in the above equation to cancel out. Also the other underlined terms, dotted and dashed accordingly, vanish. From the right-hand-side of Eq. (65), the term

$$-\frac{\Delta t}{2\Delta x} u_0 \left((\rho_\alpha\phi_\alpha)^{n+1}_{i+1} h^n_{\alpha,i+1} - (\rho_\alpha\phi_\alpha)^{n+1}_{i-1} h^n_{\alpha,i-1} \right),$$

can be written by inserting the discretization of the enthalpy (18), which is

$$h^n_\alpha := \frac{\rho^n_\alpha}{\rho^{n+1}_\alpha} \left(e^n_\alpha + \frac{p^n_\alpha}{\rho^n_\alpha} \right),$$

hence obtaining

$$-\frac{\Delta t}{2\Delta x} u_0 \left((\rho_\alpha e_\alpha)^n_{i+1} \phi^{n+1}_{\alpha,i+1} - (\rho_\alpha e_\alpha)^n_{i-1} \phi^{n+1}_{\alpha,i-1} \right).$$

In Eq. (65), substituting the above operator, we are, thus, left with

$$p_0 \frac{\phi^{n+1}_{\alpha,i}}{\gamma - 1} = p_0 \frac{\phi^n_{\alpha,i}}{\gamma - 1} - \frac{\Delta t}{2\Delta x} u_0 \left((\rho_\alpha e_\alpha)^n_{i+1} \phi^{n+1}_{\alpha,i+1} - (\rho_\alpha e_\alpha)^n_{i-1} \phi^{n+1}_{\alpha,i-1} \right) - \underline{\frac{\Delta t}{2\Delta x} u_0 p_0 \left(\phi^{n+1}_{\alpha,i+1} - \phi^{n+1}_{\alpha,i-1} \right)} + \underline{\frac{\Delta t}{2\Delta x} p_0 u_0 \left(\phi^{n+1}_{\alpha,i+1} - \phi^{n+1}_{\alpha,i-1} \right)}. \tag{66}$$

Once again, the last two underlined dashed terms cancel out. Now, expressing the internal energy as a function of pressure and density by virtue of the ideal gas EOS, one obtains

$$\begin{aligned} p_0 \frac{\phi_{\alpha,i}^{n+1}}{\gamma-1} &= p_0 \frac{\phi_{\alpha,i}^{n}}{\gamma-1} - \frac{\Delta t}{2\Delta x} u_0 \frac{p_0}{\gamma-1}\left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right) \\ &= p_0 \frac{1}{\gamma-1}\left(\phi_{\alpha,i}^{n} - \frac{\Delta t}{2\Delta x} u_0 \left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right)\right) \\ &= p_0 \frac{\phi_{\alpha,i}^{n+1}}{\gamma-1}, \end{aligned} \tag{67}$$

since the right-hand-side exactly corresponds to the fully discrete evolution of the volume fraction given by (64d). Consequently, p_0 is the solution of the pressure sub-system, which remains constant.

Finally, the update of momentum and total energy is performed as given in (32a) and (32b), respectively, with the enthalpy definition (18):

$$\begin{aligned} (\rho_\alpha \phi_\alpha u_\alpha)_i^{n+1} &= q_{\alpha,i}^{\star} - \frac{\Delta t}{2\Delta x} p_0 \left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right) + \frac{\Delta t}{2\Delta x} p_0 \left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right) \\ &= q_{\alpha,i}^{\star} = u_0 (\rho_\alpha \phi_\alpha)_i^{n+1}, \end{aligned} \tag{68a}$$

$$\begin{aligned} (\rho_\alpha \phi_\alpha E_\alpha)_i^{n+1} &= E_{\alpha,i}^{\star} - \frac{\Delta t}{2\Delta x} u_0 \left(\frac{p_0}{\gamma-1} + p_0\right)\left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right) + \frac{\Delta t}{2\Delta x} p_0 u_0 \left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right) \\ &= \frac{p_0}{\gamma-1}\left(\phi_{\alpha,i}^{n} - \frac{\Delta t}{2\Delta x} u_0 \left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right)\right) + \frac{u_0^2}{2}(\rho_\alpha \phi_\alpha)_i^{n+1} \\ &= \frac{p_0}{\gamma-1}\phi_{\alpha,i}^{n+1} + \frac{u_0^2}{2}(\rho_\alpha \phi_\alpha)_i^{n+1}. \end{aligned} \tag{68b}$$

The fully discrete scheme presented is, therefore, well-balanced in the sense that it can exactly preserve the moving equilibrium originating from the initial condition (63). Numerical evidence supporting this property is provided by the test case RP0 discussed in Section 4.2.

Well-balanced discretization property with numerical dissipation. The well-balanced property is respected even in the presence of the additional numerical stabilization terms in the volume fraction equation. In this case, the maximum convective eigenvalue according to (27) reduces to the constant velocity, i.e. $a_\Omega^n = u_0$, and thus the viscosity coefficient (37) simplifies to

$$\nu^{\pm} = \frac{1}{2}\Delta x\, u_0.$$

Consequently, the update of the volume fraction Eq. (36) writes

$$\phi_{s,i}^{n+1} = \phi_{s,i}^{n} - \frac{\Delta t}{2\Delta x} u_0 \left(\phi_{s,i+1}^{n+1} - \phi_{s,i-1}^{n+1}\right) + \frac{\Delta t}{2\Delta x} u_0 \left(\phi_{s,i+1}^{n} - 2\phi_{s,i}^{n} + \phi_{s,i-1}^{n}\right). \tag{69}$$

The stabilization term in the above equation must be added to the right hand side of the wave equation for the pressure (65), so that Eq. (67) now becomes

$$\begin{aligned} p_0 \frac{\phi_{\alpha,i}^{n+1}}{\gamma-1} &= p_0 \frac{\phi_{\alpha,i}^{n}}{\gamma-1} - \frac{\Delta t}{2\Delta x} u_0 \frac{p_0}{\gamma-1}\left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right) + \frac{\Delta t}{2\Delta x}\frac{p_0}{\gamma-1} u_0 \left(\phi_{s,i+1}^{n} - 2\phi_{s,i}^{n} + \phi_{s,i-1}^{n}\right) \\ &= p_0 \frac{1}{\gamma-1}\left(\phi_{\alpha,i}^{n} - \frac{\Delta t}{2\Delta x} u_0 \left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right) + \frac{\Delta t}{2\Delta x} u_0 \left(\phi_{s,i+1}^{n} - 2\phi_{s,i}^{n} + \phi_{s,i-1}^{n}\right)\right) \\ &= p_0 \frac{\phi_{\alpha,i}^{n+1}}{\gamma-1}, \end{aligned} \tag{70}$$

where the numerical dissipation has been rescaled in order to match the correct units by the factor

$$\frac{\partial(\rho_\alpha \phi_\alpha e_\alpha)_i^n}{\partial \phi_{\alpha,i}^n} = \frac{\partial}{\partial \phi_{\alpha,i}^n}\left(\frac{(\phi_\alpha p_\alpha)_i^n}{\gamma_\alpha - 1}\right) = \frac{p_0}{\gamma_\alpha - 1}. \tag{71}$$

Finally, the update of the total energy (68b) is also modified by adding the stabilization term used in (69) properly rescaled by the factor (71), hence leading to

$$\begin{aligned} (\rho_\alpha \phi_\alpha E_\alpha)_i^{n+1} &= E_{\alpha,i}^{\star} - \frac{\Delta t}{2\Delta x} u_0 \left(\frac{p_0}{\gamma-1} + p_0\right)\left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right) + \frac{\Delta t}{2\Delta x} p_0 u_0 \left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right) \\ &\quad + \frac{\Delta t}{2\Delta x}\frac{p_0}{\gamma-1} u_0 \left(\phi_{s,i+1}^{n} - 2\phi_{s,i}^{n} + \phi_{s,i-1}^{n}\right) \\ &= \frac{p_0}{\gamma-1}\left(\phi_{\alpha,i}^{n} - \frac{\Delta t}{2\Delta x} u_0 \left(\phi_{\alpha,i+1}^{n+1} - \phi_{\alpha,i-1}^{n+1}\right) + \frac{\Delta t}{2\Delta x} u_0 \left(\phi_{s,i+1}^{n} - 2\phi_{s,i}^{n} + \phi_{s,i-1}^{n}\right)\right) + \frac{u_0^2}{2}(\rho_\alpha \phi_\alpha)_i^{n+1} \\ &= \frac{p_0}{\gamma-1}\phi_{\alpha,i}^{n+1} + \frac{u_0^2}{2}(\rho_\alpha \phi_\alpha)_i^{n+1}. \end{aligned} \tag{72}$$

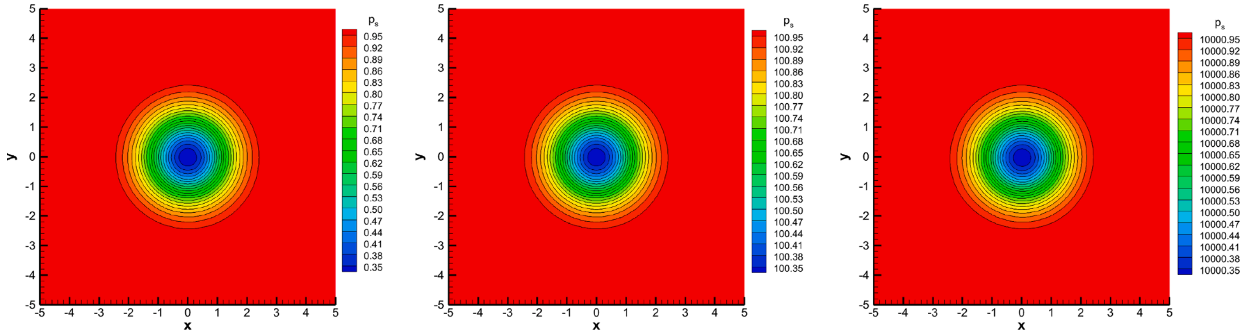

Fig. 1. Smooth vortex problem at time $t_f = 5$, on a mesh with characteristic size $\varrho = 1/16$. Numerical distribution of the pressure of the solid phase, with a background pressure of $p_0 = 1$ (left), $p_0 = 10^2$ (middle), and $p_0 = 10^4$ (right).

4. Numerical results

This section presents a suite of numerical test cases for the Baer–Nunziato model that aim at validating the novel semi-implicit finite volume scheme in terms of accuracy and robustness. The new scheme is labeled with SIFV-BN. Different flow regimes are examined, to show the versatility of the method in the presence of different Mach numbers ranging in the interval $[10^{-3}, 10^1]$. The time step is always computed according to the CFL-type stability condition (11) with CFL $= 0.9$. If the initial flow speed is zero, only the first time step is given by (6) so that it is limited by the sound speed. If not stated otherwise, the dimensionless form of the governing equations is used.

4.1. Numerical convergence study

For the convergence study, a modified version of the smooth vortex problem proposed in Hu and Shu [66] is chosen, following the setup of Dumbser et al. [67]. All the details related to the definition of the initial condition for this test problem can be found in Dumbser et al. [67], therefore only a qualitative description is provided hereafter. An isentropic vortex is superimposed to a constant background state on the computational domain $\Omega = [-5, 5] \times [-5, 5]$, with periodic boundary conditions. The different phases are characterized by the ideal gas EOS, with $\gamma_f = 1.35$, and $\gamma_s = 1.4$. The exact solution of this problem is the passive convection of the vortex with the mean advection velocity $\mathbf{v}_c = (2, 2)$, which is not aligned with the mesh directions. At the final time, $t_f = 5$, the vortex has done a complete revolution and is back to its initial location. The test is conducted on a series of successively refined computational meshes. Moreover, to study the convergence of the scheme at different Mach numbers, different background pressures, $p_0 \in [1, 10^4]$, are imposed to define the initial condition. Fig. 1 shows the vortex at the final time of the simulation, at different Mach numbers on the same mesh, demonstrating that the solution keeps the same accuracy, and no spurious oscillations arise. The new all Mach solver ensures that the solution remains independent of the Mach number, preserving its vortical structure with no degradation. The errors are measured in L_2 norm with respect to the analytical solution given in Dumbser et al. [67], for density, x−momentum, and volume fraction of the solid phase. The results are reported in Table 1, demonstrating both the accuracy and the asymptotic preserving property of the novel SIFV-BN scheme. The second order of accuracy is still conserved for low Mach numbers.

4.2. Riemann problems

In one space dimension, several Riemann problems, for which the exact solution is known [68–70], are tested. The one-dimensional domain is $\Omega = [0, 1]$, which is discretized by $N_x = 1000$ cells, with Dirichlet boundary conditions. The initial discontinuity is located at $x_D = 0.5$. The initial conditions for the left (L) and right (R) states are listed in Table 2, together with the EOS parameters of the two phases, the source term coefficients, and the final time of each Riemann Problem (RP). Figs. 2–9 depict a comparison between the numerical and the reference solution, that is computed using an explicit second order TVD finite volume scheme on a fine mesh composed of 10’000 cells. The first test, RP0, is actually the implementation of a moving contact discontinuity, to numerically demonstrate the well-balanced property proven in Section 3.7. Fig. 2 shows the exact preservation of the constant pressure and velocity for both phases, while the jumps in density and volume fraction are captured. RP1, RP2, RP3, and RP4 are shock tube problems with different EOS parameters, involving higher difference in the variable jumps or the initial velocity. In Figs. 3–6, the rarefaction waves, and in particular shock waves, are nicely captured, with a slightly increased difficulty for RP3 and RP4. The numerical scheme exhibits small oscillations for RP5 and RP6 (Figs. 7 and 8), but the agreement of the numerical solution with the reference is still very good. Finally, in RP7, the classical Sod shock tube, the source terms are present, both for pressure (which is stiff) and velocity. The exact solution, in this case, is obtained using an exact Riemann solver for the compressible Euler equations, with different polytropic indexes on the left and right side of the contact discontinuity. Also in this case, the numerical scheme is able to provide very satisfying results, see Fig. 9. This collection of test cases shows the ability of our method to comply with the high Mach regime, although it is not specifically designed for simulating sharp discontinuities.

Table 1
Numerical convergence results for the smooth vortex problem, for different background pressure p_0. The errors for density, x−momentum, and volume fraction of the solid phase are given in L_2 norm at the final time $t_f = 5$.

N_x	ϱ	$L_2(\rho_s\phi_s)$	$\mathcal{O}(\rho_s\phi_s)$	$L_2(\rho_s\phi_s u_s)$	$\mathcal{O}(\rho_s\phi_s u_s)$	$L_2(\phi_s)$	$\mathcal{O}(\phi_s)$
		$p_0 = 1$					
20	0.5	4.2250E−01	–	1.0690E+00	–	1.6308E−01	–
40	0.25	1.6787E−01	1.33	3.4438E−01	1.63	6.3713E−02	1.36
80	0.125	4.6463E−02	1.85	8.7884E−02	1.97	1.6970E−02	1.91
160	0.0625	1.1957E−02	1.96	2.5092E−02	1.81	4.2119E−03	2.01
		$p_0 = 10^1$					
20	0.5	3.5478E−02	–	1.0726E+00	–	1.6402E−01	–
40	0.25	1.3797E−02	1.36	3.5107E−01	1.61	6.3683E−02	1.36
80	0.125	3.2493E−03	2.09	8.8656E−02	1.99	1.6530E−02	1.95
160	0.0625	7.6859E−04	2.08	2.5126E−02	1.82	4.0399E−03	2.03
		$p_0 = 10^2$					
20	0.5	4.9215E−03	–	1.0725E+00	–	1.6412E−01	–
40	0.25	1.6694E−03	1.56	3.5247E−01	1.61	6.3743E−02	1.36
80	0.125	4.2384E−04	1.98	8.9128E−02	1.98	1.6527E−02	1.95
160	0.0625	1.0384E−04	2.03	2.5228E−02	1.82	4.0357E−03	2.03
		$p_0 = 10^3$					
20	0.5	2.8493E−03	–	1.0675E+00	–	1.6386E−01	–
40	0.25	1.2504E−03	1.19	3.4870E−01	1.61	6.3264E−02	1.37
80	0.125	3.3387E−04	1.90	8.7773E−02	1.99	1.6292E−02	1.96
160	0.0625	8.2761E−05	2.01	2.4899E−02	1.82	3.9670E−03	2.04
		$p_0 = 10^4$					
20	0.5	9.4160E−03	–	1.0305E+00	–	1.6343E−01	–
40	0.25	4.2722E−03	1.14	3.2406E−01	1.67	6.2452E−02	1.39
80	0.125	9.1133E−04	2.23	7.9132E−02	2.03	1.5855E−02	1.98
160	0.0625	1.4772E−04	2.63	2.2924E−02	1.79	3.8419E−03	2.05

Table 2
Initial condition for the left (L) and right (R) states of the Riemann problems. The values for the EOS of the two phases are reported as well as the source terms parameters (δ,μ) and the final time.

	ρ_f	$\mathbf{u}_f$	p_f	ρ_s	$\mathbf{u}_s$	p_s	ϕ_s	t_f
RP0		$\gamma_f = 1.4,\ \pi_f = 0$			$\gamma_s = 1.4,\ \pi_s = 0$		$\delta = 0,\ \mu = 0$	
L	1.0	1.0	1.0	1.0	1.0	1.0	0.8	0.2
R	0.5	1.0	1.0	0.5	1.0	1.0	0.4	
RP1 [69]		$\gamma_f = 1.4,\ \pi_f = 0$			$\gamma_s = 1.4,\ \pi_s = 0$		$\delta = 0,\ \mu = 0$	
L	0.5	0.0	1.0	1.0	0.0	1.0	0.4	0.1
R	1.5	0.0	2.0	2.0	0.0	2.0	0.8	
RP2 [69]		$\gamma_f = 1.4,\ \pi_f = 0$			$\gamma_s = 3.0,\ \pi_s = 100$		$\delta = 0,\ \mu = 0$	
L	1.5	0.0	2.0	800.0	0.0	500.0	0.4	0.1
R	1.0	0.0	1.0	1000.0	0.0	600.0	0.3	
RP3 [69]		$\gamma_f = 1.4,\ \pi_f = 0$			$\gamma_s = 1.4,\ \pi_s = 0$		$\delta = 0,\ \mu = 0$	
L	1.0	0.0	1.0	1.0	0.9	2.5	0.9	0.1
R	1.2	1.0	2.0	1.0	0.0	1.0	0.2	
RP4 [70]		$\gamma_f = 1.35,\ \pi_f = 0$			$\gamma_s = 3.0,\ \pi_s = 3400$		$\delta = 0,\ \mu = 0$	
L	2.0	0.0	3.0	1900.0	0.0	10.0	0.2	0.15
R	1.0	0.0	1.0	1950.0	0.0	1000.0	0.9	
RP5 [70]		$\gamma_f = 1.4,\ \pi_f = 0$			$\gamma_s = 1.4,\ \pi_s = 0$		$\delta = 0,\ \mu = 0$	
L	0.2	0.0	0.3	1.0	0.0	1.0	0.8	0.2
R	1.0	0.0	1.0	1.0	0.0	1.0	0.3	
RP6 [68]		$\gamma_f = 1.4,\ \pi_f = 0$			$\gamma_s = 1.4,\ \pi_s = 0$		$\delta = 0,\ \mu = 0$	
L	0.5806	1.5833	1.375	0.2068	1.4166	0.0416	0.1	0.1
R	0.4890	-0.70138	0.986	2.2263	0.9366	6.0	0.2	
RP7 [54]		$\gamma_f = 1.67,\ \pi_f = 0$			$\gamma_s = 1.4,\ \pi_s = 0$		$\delta = 10^3,\ \mu = 10^2$	
L	1.0	0.0	1.0	1.0	0.0	1.0	0.99	0.2
R	0.125	0.0	0.1	0.125	0.0	0.1	0.01	

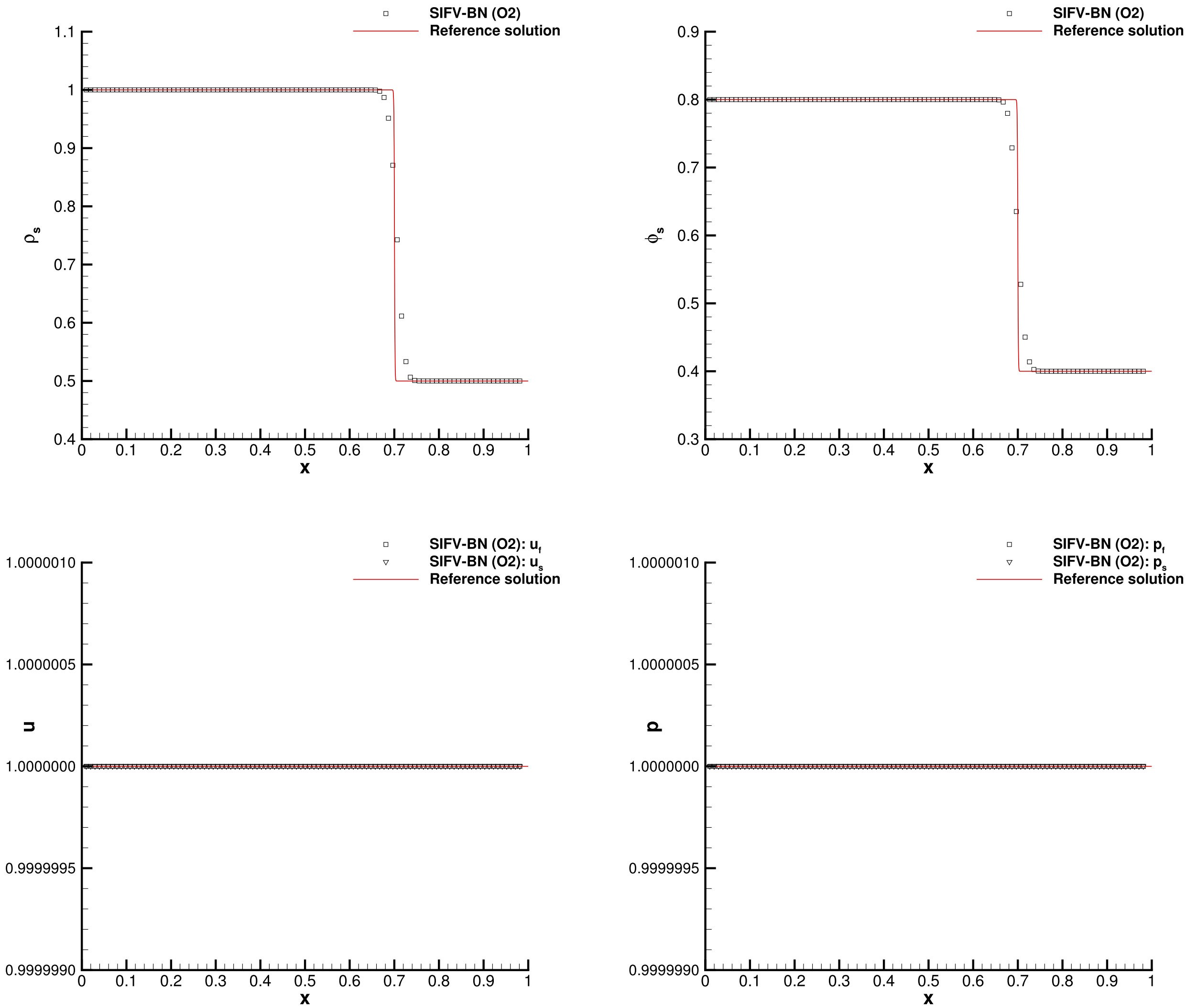

Fig. 2. Riemann problem RP0 at time $t_f = 0.2$. Numerical results for density, volume fraction, horizontal velocity and pressure (from top left to bottom right) compared against the reference solution.

Explosion problem with and without inter-phase drag relaxation. To better investigate the influence of the relaxation terms, we solve a cylindrical explosion problem on the domain $\Omega = [-1, 1]^2$ discretized with $N_x \times N_y = 1000 \times 1000$ cells. The initial condition is given by the left and the right state of RP2 in Table 2 separated by a circular discontinuity located at radius $R = 0.4$. We run the simulation until time $t_f = 0.15$ with and without inter-phase drag relaxation terms, namely setting $\delta = 10^5$ and $\delta = 0$. The reference solution is computed relying on an explicit second order TVD finite volume scheme on a fine mesh composed of 10'000 cells. The results are depicted in Fig. 10, showing an excellent agreement with the reference solution. The effect of the stiff source terms is particularly evident in the phase velocities, which relax to the mixture velocity when $\delta = 10^5$. The cylindrical symmetry of the solution can be also qualitatively appreciated looking at the three-dimensional distribution of the fluid phase density. A comparison in terms of computational efficiency against a fully explicit scheme is reported in Table 5 for the case $\delta = 10^5$, showing a gaining factor of 43. This is obviously due to the presence of the stiff drag relaxation coefficient, which is implicitly solved in our numerical method.

4.3. Two-dimensional Riemann problems

Increasing the spatial dimension does not change the methodology, since we use Cartesian grids, but it increases the burden of the computational cost. The novel numerical scheme is parallelized using MPI, allowing simulations with a very high number of degrees of freedom. Here, we run a couple of genuinely two-dimensional Riemann problems proposed in Dumbser and Boscheri [54], which are inspired from Zhang and Zheng [71], Kurganov and Tadmor [72]. The initial condition is defined by four piecewise constant states, prescribed on each of the four quadrants of the squared domain $\Omega = [-0.5, 0.5] \times [-0.5, 0.5]$. Reflective wall boundary conditions are applied everywhere. The initial conditions and the parameters for the two test cases are summarized in Table 3. The first Riemann

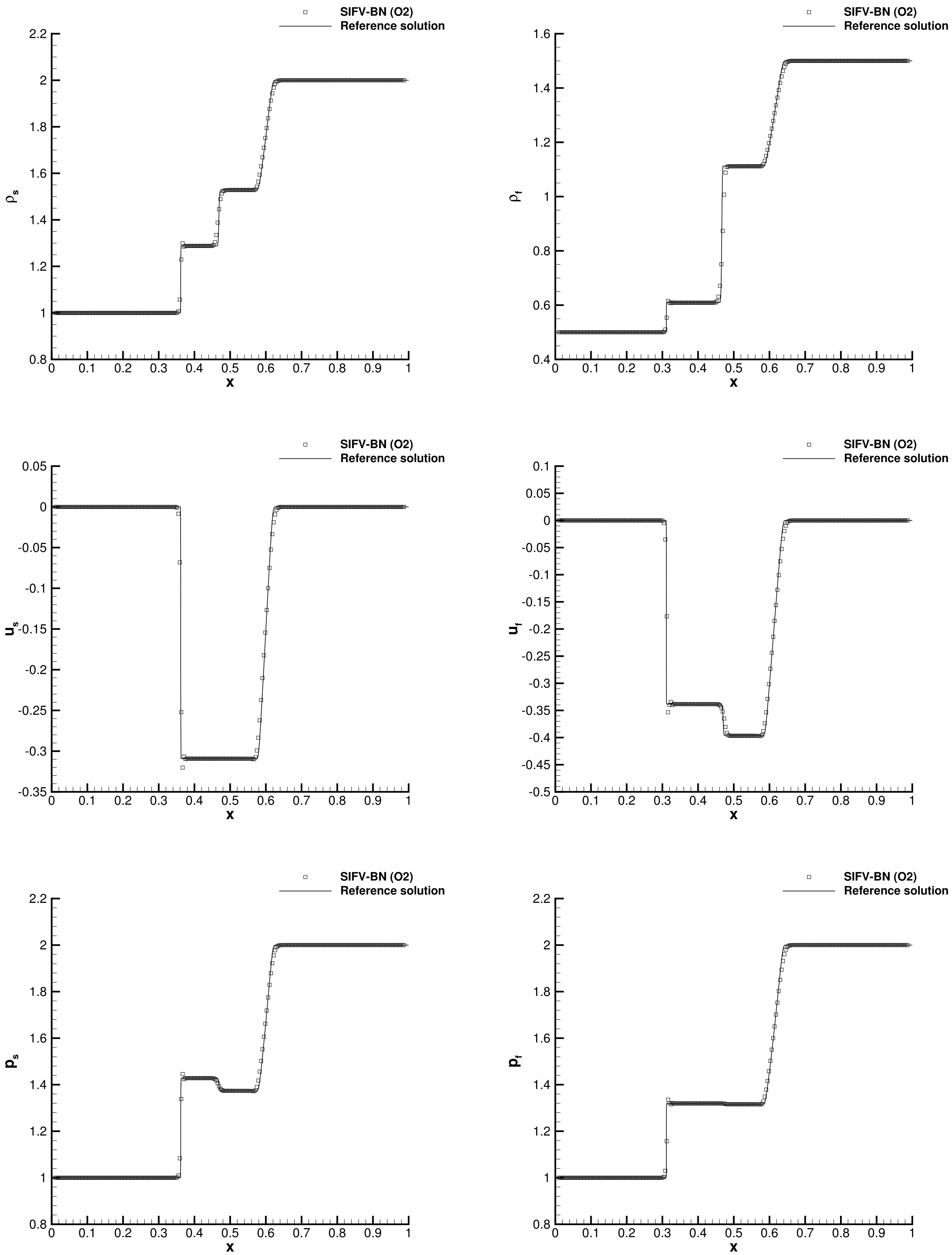

Fig. 3. Riemann problem RP1 at time $t_f = 0.1$. Numerical results for density (top), horizontal velocity (middle) and pressure (bottom) compared against the reference solution. Left: solid phase. Right: fluid phase.

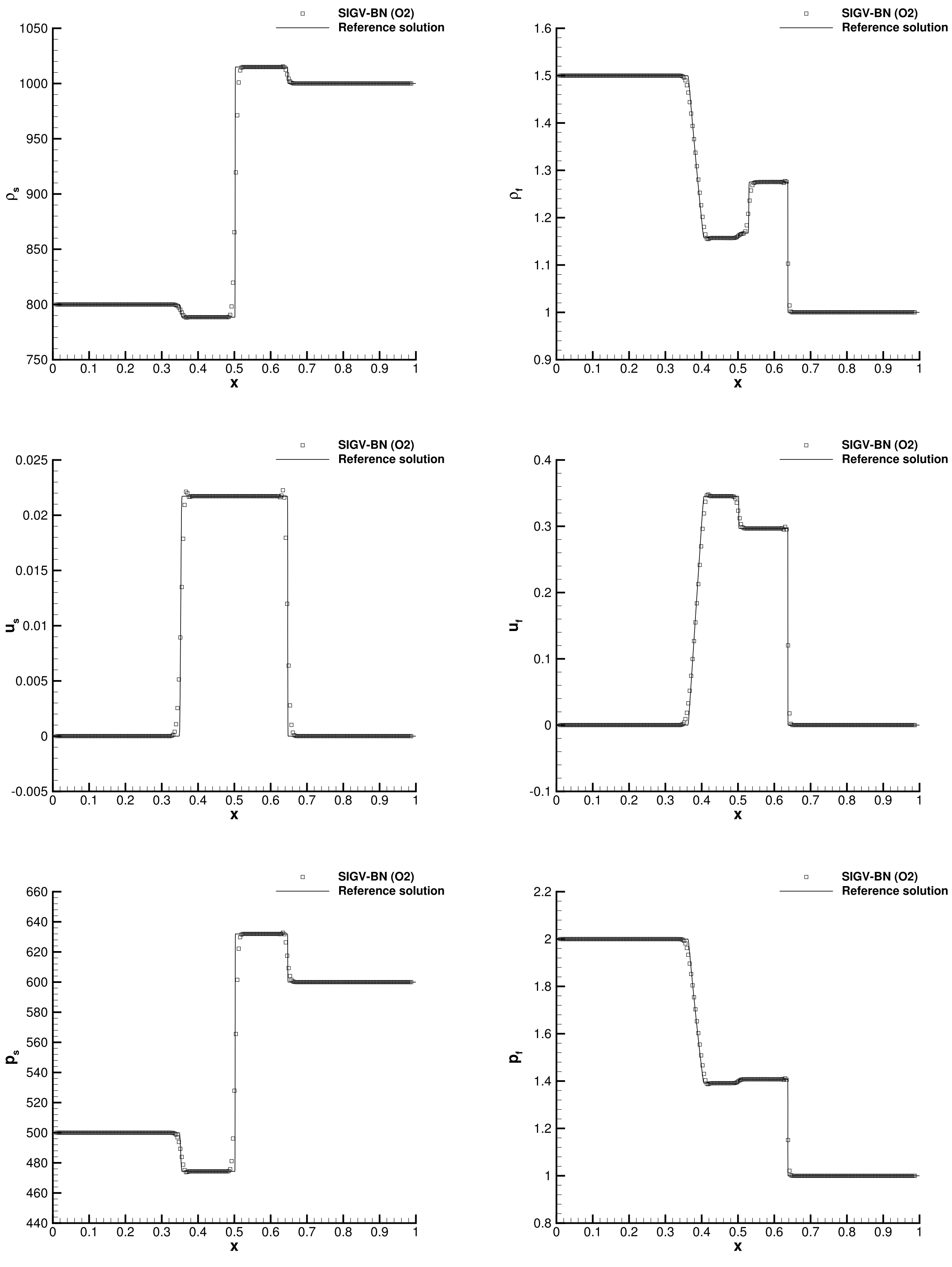

Fig. 4. Riemann problem RP2 at time $t_f = 0.1$. Numerical results for density (top), horizontal velocity (middle) and pressure (bottom) compared against the reference solution. Left: solid phase. Right: fluid phase.

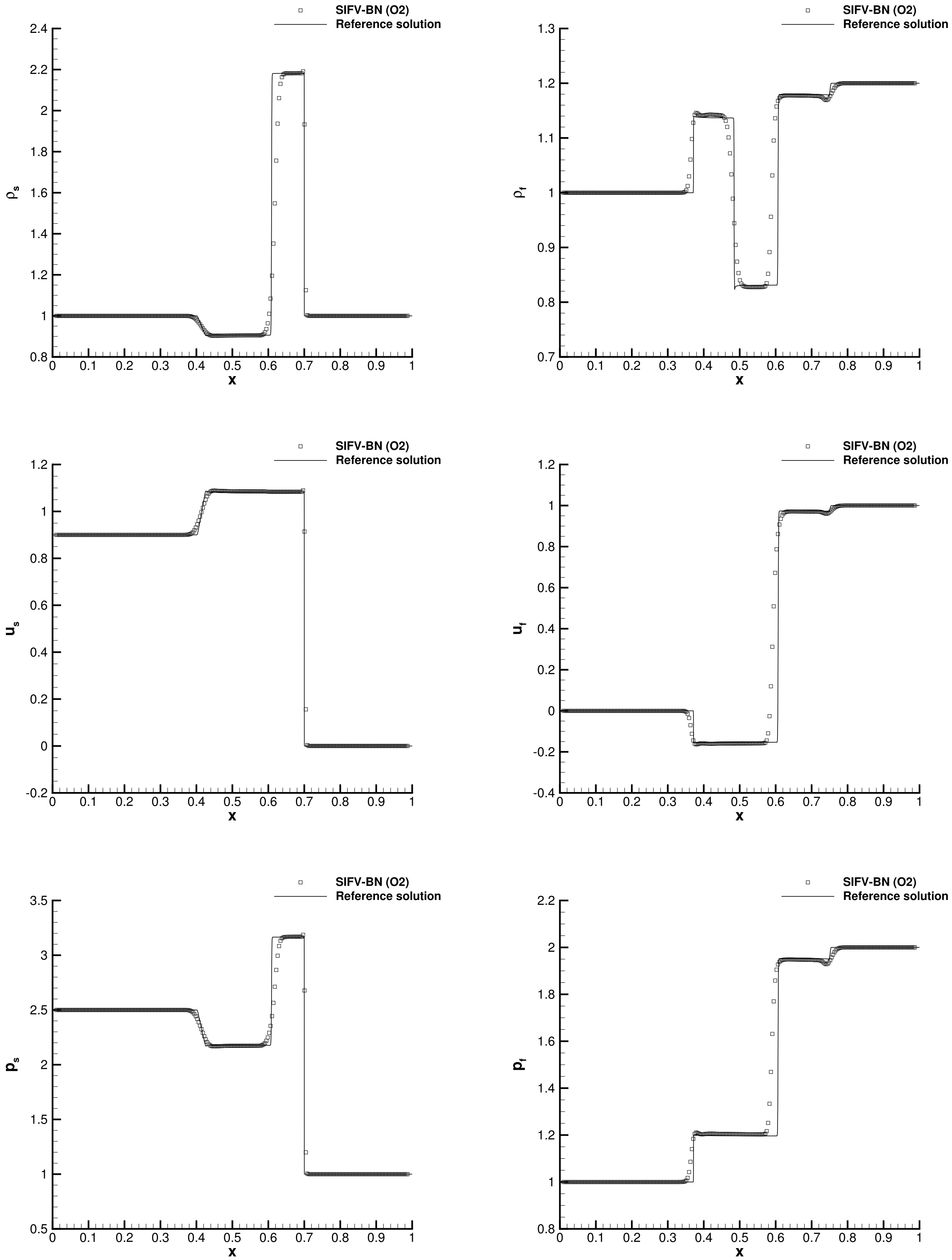

Fig. 5. Riemann problem RP3 at time $t_f = 0.1$. Numerical results for density (top), horizontal velocity (middle) and pressure (bottom) compared against the reference solution. Left: solid phase. Right: fluid phase.

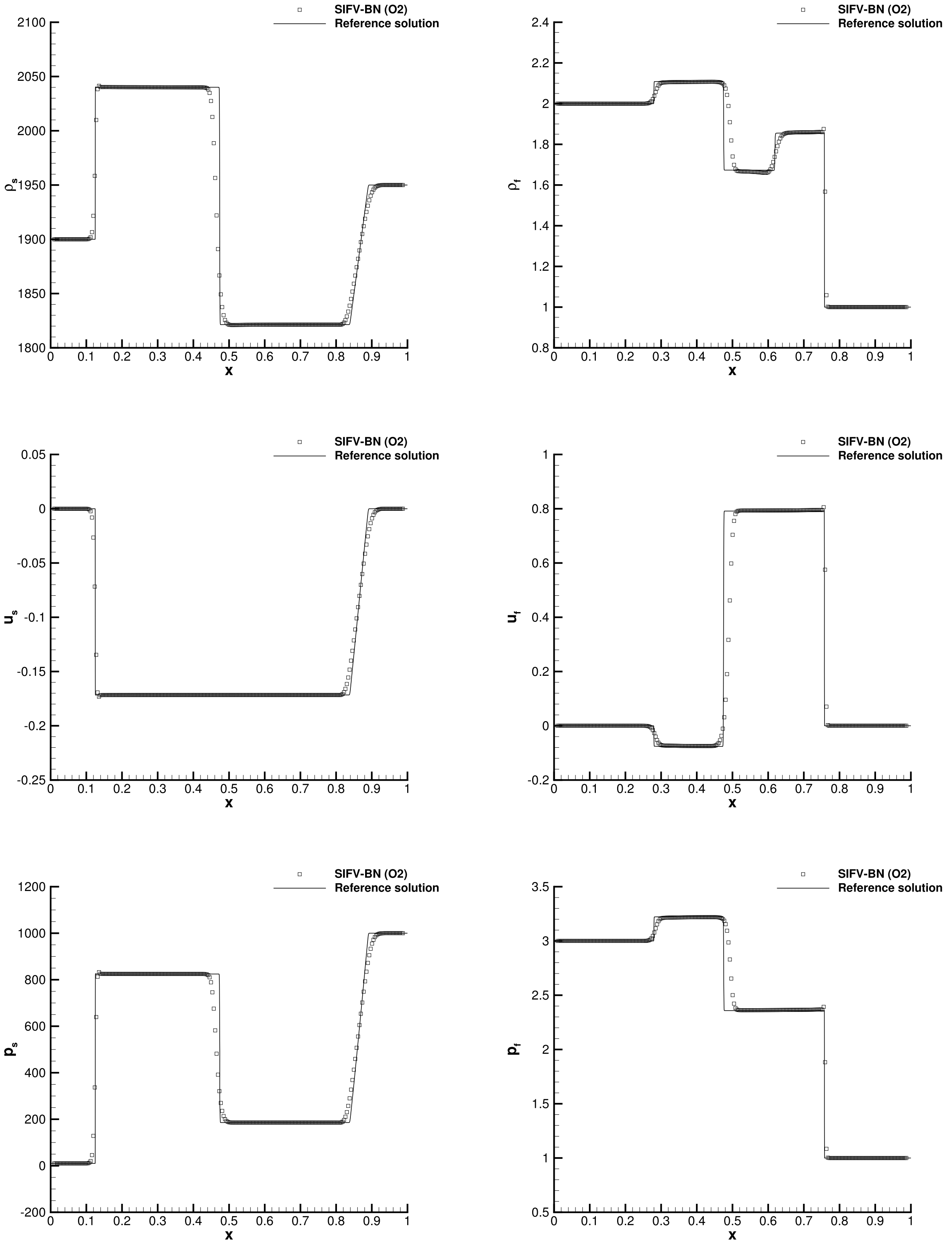

Fig. 6. Riemann problem RP4 at time $t_f = 0.15$. Numerical results for density (top), horizontal velocity (middle) and pressure (bottom) compared against the reference solution. Left: solid phase. Right: fluid phase.

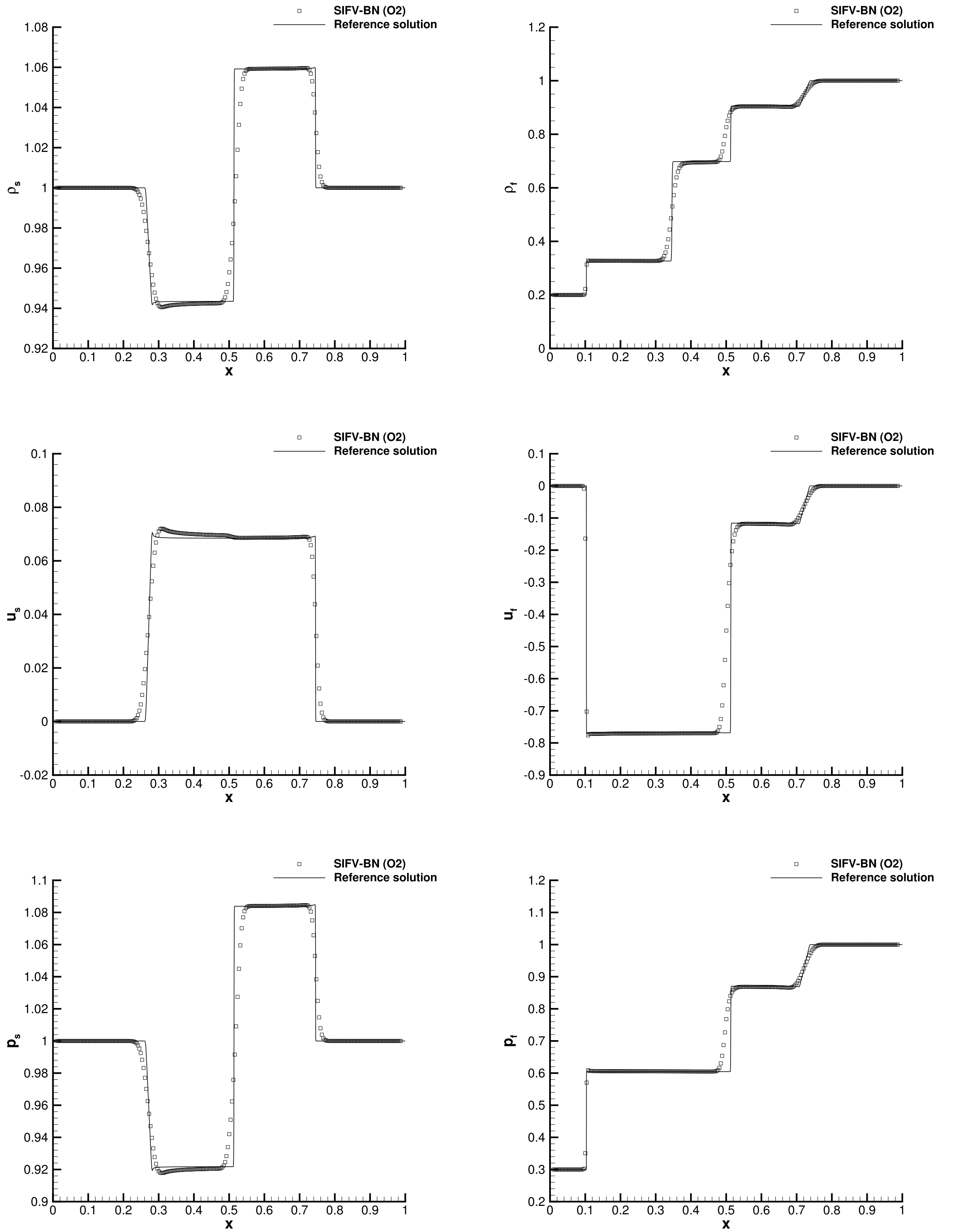

Fig. 7. Riemann problem RP5 at time $t_f = 0.2$. Numerical results for density (top), horizontal velocity (middle) and pressure (bottom) compared against the reference solution. Left: solid phase. Right: fluid phase.

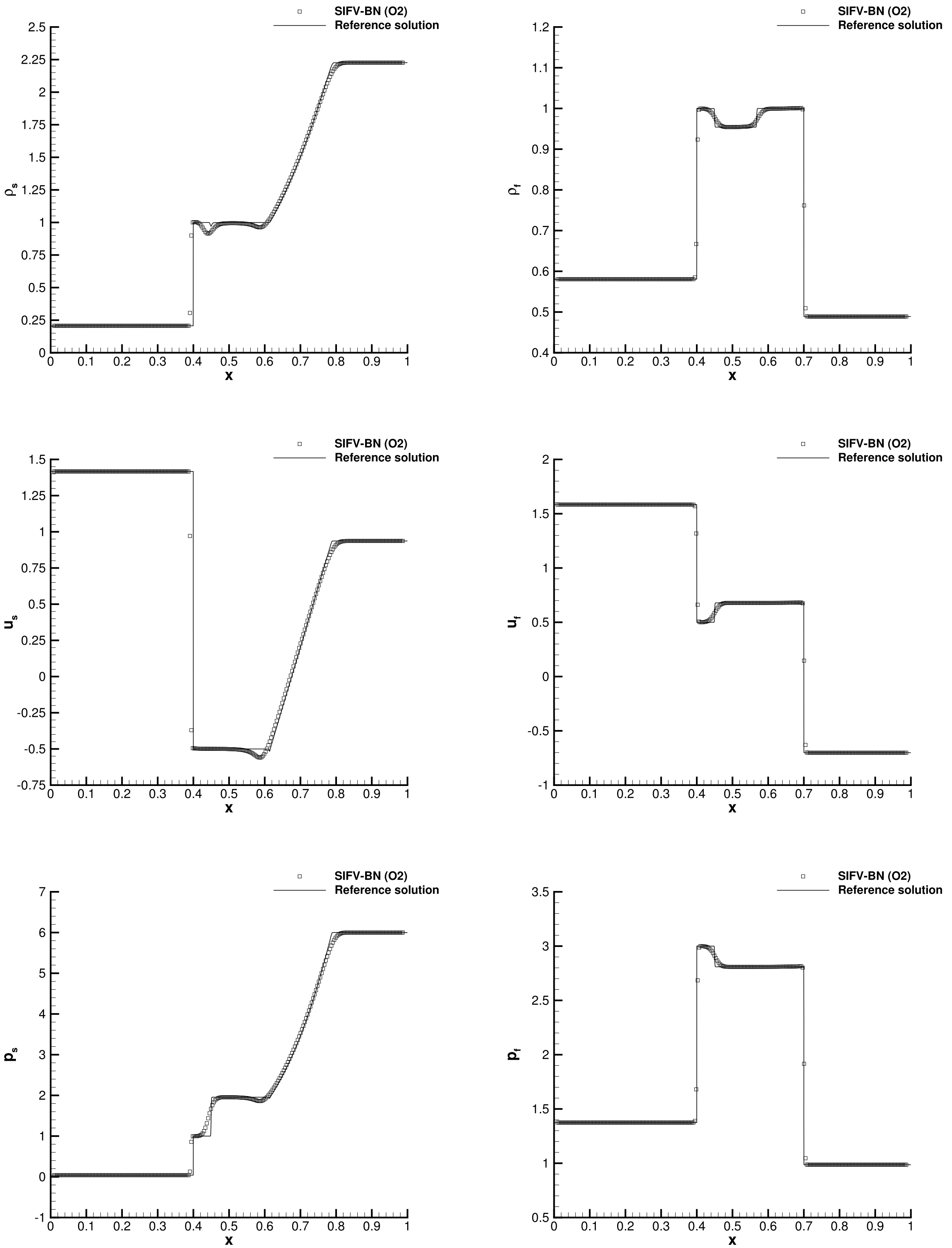

Fig. 8. Riemann problem RP6 at time $t_f = 0.1$. Numerical results for density (top), horizontal velocity (middle) and pressure (bottom) compared against the reference solution. Left: solid phase. Right: fluid phase.

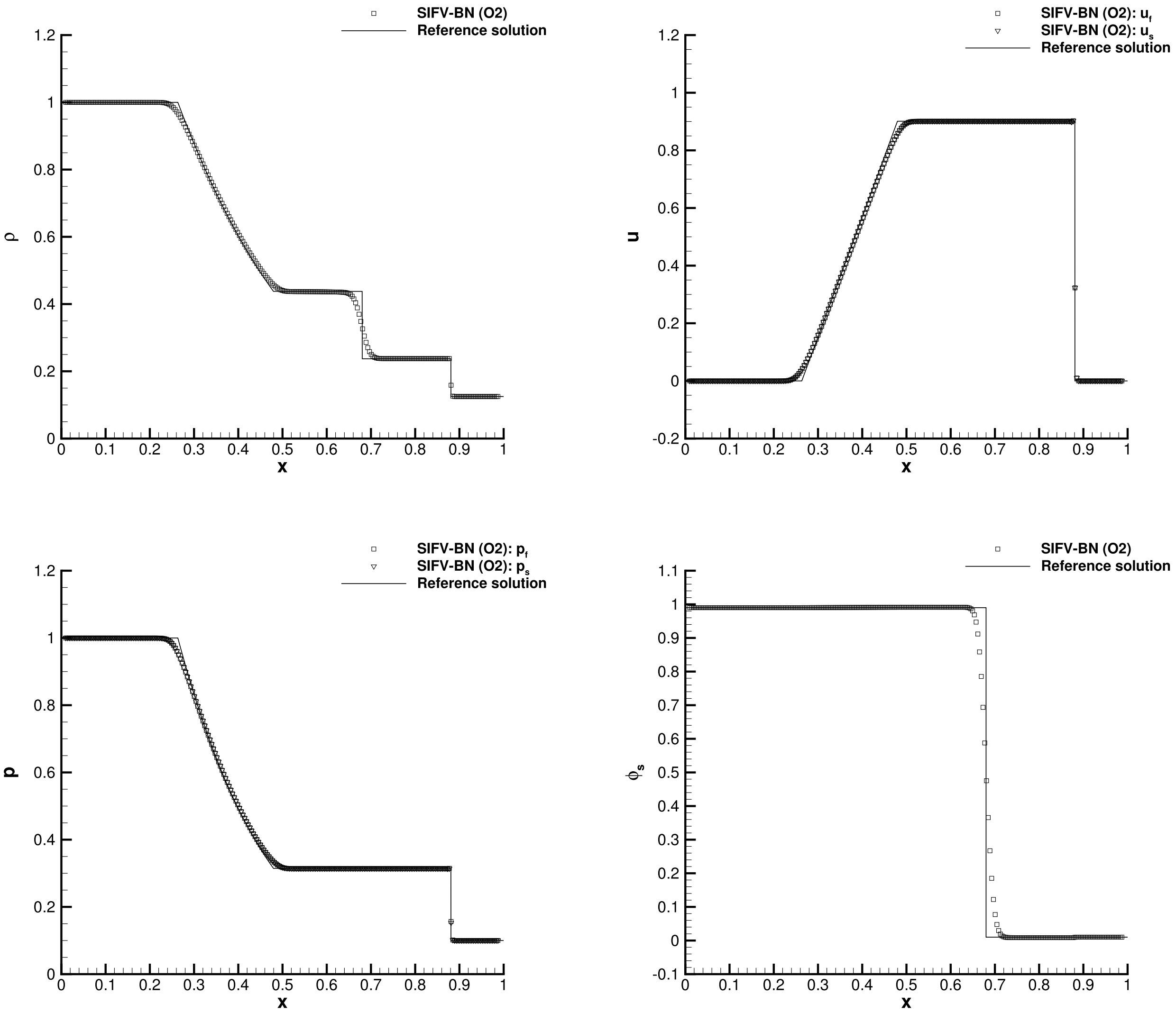

Fig. 9. Riemann problem RP7 at time $t_f = 0.2$. Numerical results for mixture density, horizontal velocity, pressure and volume fraction (from top left to bottom right) compared against the reference solution.

problem involves stiff relaxation sources, while the second one only considers the homogeneous system. The reference solution is computed with a high order ADER scheme on a very fine mesh composed of 2’277’668 triangles, with characteristic mesh spacing $h = 1/1000$, see Dumbser et al. [67], Toro et al. [73].

The numerical results obtained with the SIFV-BN scheme are displayed in the left column of Figs. 11 and 12, and compared to the reference solution on the right column, for the density of both phases and the volume fraction. The results are strikingly in accordance with the reference from the qualitative viewpoint.

4.4. 2D Taylor–Green vortex at low Mach number

The two-dimensional version of the original Taylor–Green vortex [74] is used here to perform a numerical convergence study in the low Mach regime. The analytical solution for a single-phase fluid in the incompressible limit reads

$$\begin{aligned} \rho &= \rho_0, \\ u &= \sin(x)\cos(y), \\ v &= -\cos(x)\sin(y), \\ p &= p_0 + \frac{1}{4}(\cos(2x) + \cos(2y)). \end{aligned} \tag{73}$$

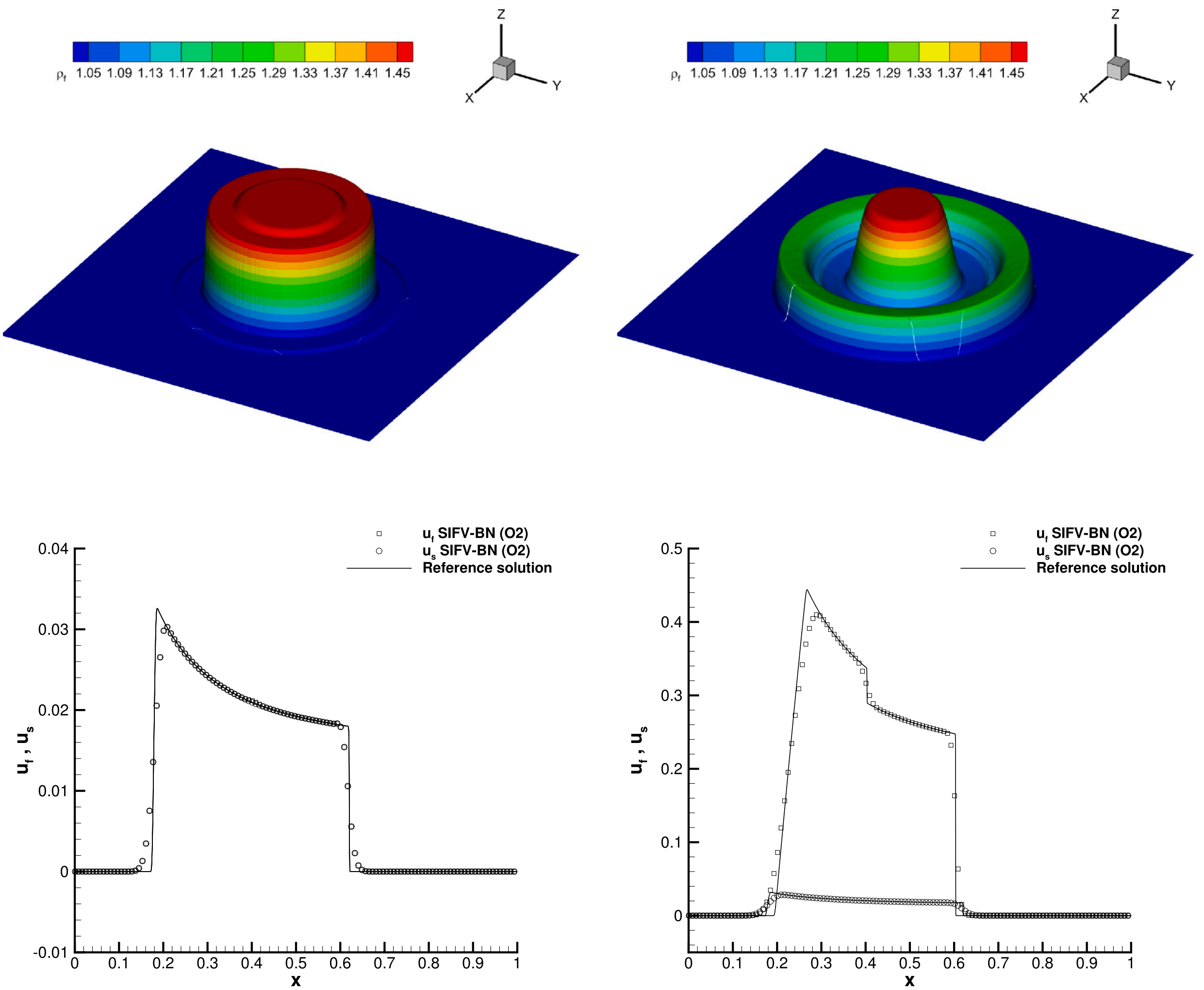

Fig. 10. Explosion problem at time $t_f = 0.15$ with (left column) and without (right column) inter-phase drag relaxation $\delta = 10^5$. Numerical results for the fluid density (top row) and for the horizontal velocity of both phases with comparison against the reference solution (bottom row)..

The computational domain is $\Omega = [0, 2\pi]^2$ with periodic boundaries, and it is discretized with a mesh composed of $N_x \times N_y = 50 \times 50$ cells. This flow is analytically divergence-free, and the objective is to verify that the numerical scheme achieves a discrete divergence-free condition as the Mach number goes to zero with second order of convergence. For this purpose, we set $\rho_0 = 1$ and we assign the initial condition (73) to each phase. We adopt the ideal gas EOS with $\gamma_f = \gamma_s = 1.4$. This test case is run on a sequence of successively diminishing Mach number regimes, by increasing the background pressure p_0, until a final time of $t_f = 0.2$, using a constant time step of $\Delta t = 2.5 \cdot 10^{-3}$. For retrieving a single-phase behavior, the volume fraction is set to $\phi_s = 0.5$, while we impose $\phi_s = 0.5 + 0.1\,(\cos(2x) + \cos(2y))$ to allow interactions between the two phases. The error in the velocity divergence for the mixture velocity $\nabla \cdot \mathbf{u} = 0$ is computed in the L_1, L_2, and L_∞ norms using the discrete cell-centered divergence operator

$$\nabla_h \cdot \mathbf{u} = \frac{u_{i+1} - u_{i-1}}{2\Delta x} + \frac{v_{i+1} - v_{i-1}}{2\Delta y}.$$

The convergence rates and the errors are reported in Table 4, showing that the scheme achieves second order Mach convergence for Mach numbers up to $M = 10^{-3}$, hence imposing a background pressure up to $p_0 = 10^6$. Fig. 13 shows the distribution of the divergence of the mixture velocity field for $M = 10^{-3}$, confirming that the scheme approaches the divergence-free condition with second order of convergence, up to errors of order 10^{-7}. One-dimensional cuts of the numerical solution are also presented and compared to the reference solution. It is worth noting that for even lower Mach numbers the scheme does not converge, as recently observed in Orlando et al. [75]. This is due to the cell-centered spatial discretization adopted in our numerical scheme, that is not a divergence preserving operator. Future research will concern the development of div-curl preserving schemes, such as those presented in Boscheri et al. [59], in the context of multi-phase flows.

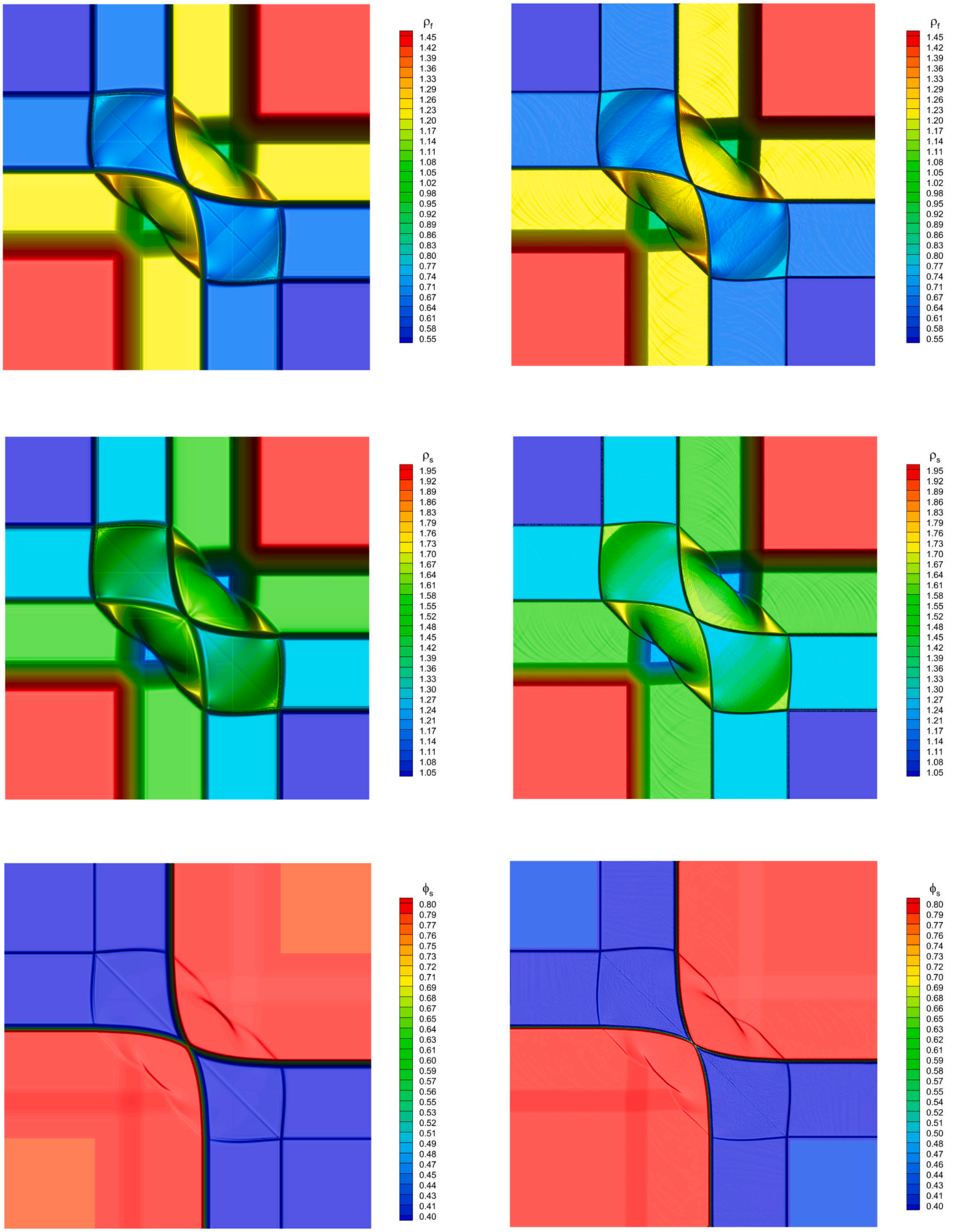

Fig. 11. 2D Riemann problem C1 at time $t_f = 0.15$. Numerical results for solid density, fluid density and volume fraction (from top to bottom). Left: second order SIFV-BN scheme. Right: reference solution computed with a third order ADER-WENO finite volume scheme [67,73].

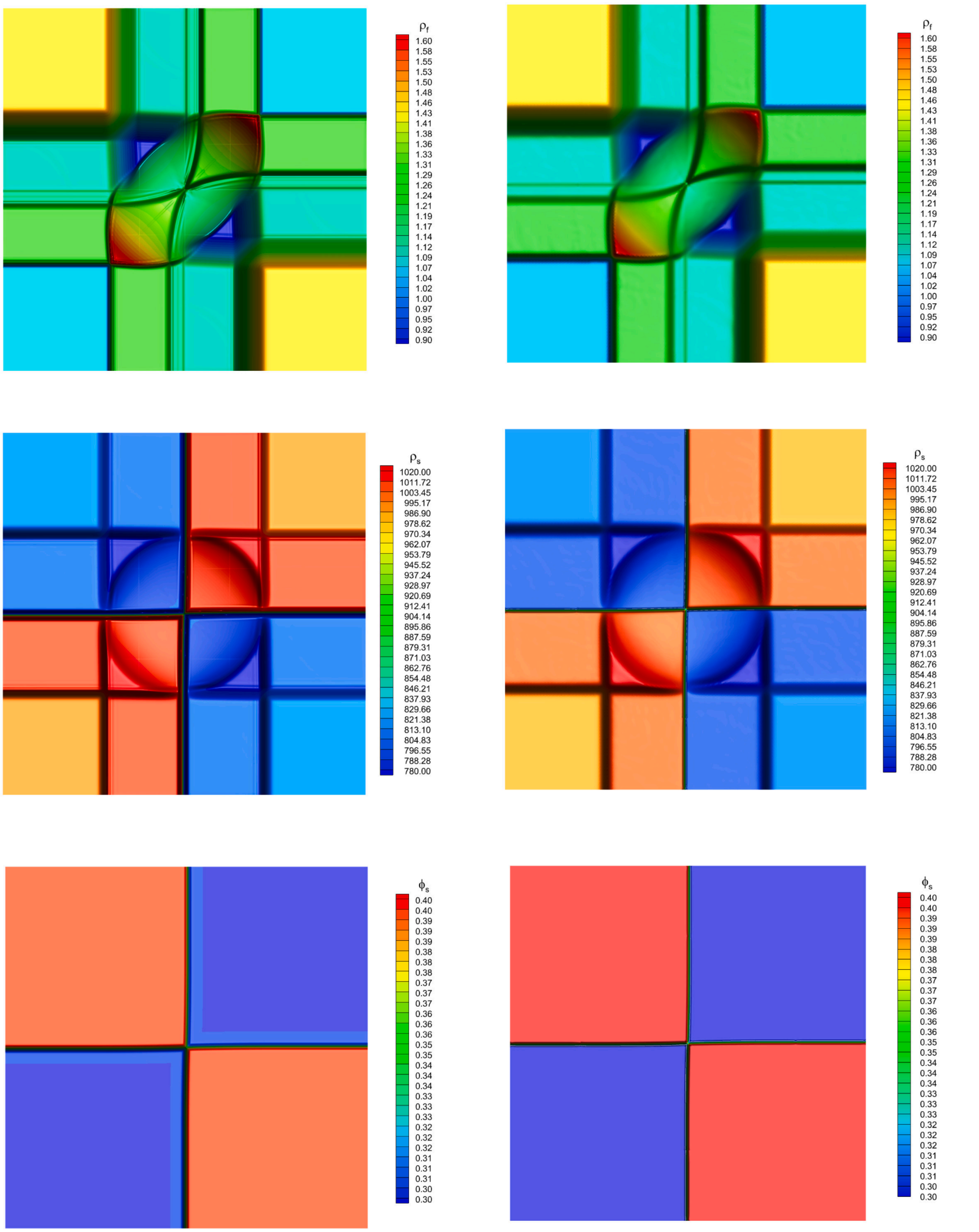

Fig. 12. 2D Riemann problem C2 at time $t_f = 0.15$. Numerical results for solid density, fluid density and volume fraction (from top to bottom). Left: second order SIFV-BN scheme. Right: reference solution computed with a third order ADER-WENO finite volume scheme [67,73].

Table 3
Initial condition for the top right (TR), top left (TL), bottom right (BR), and bottom left (BL) quadrant states of the two-dimensional Riemann problems. The values for the EOS of the two phases are given as well as the source terms parameters and the final time.

	ρ_f	$\mathbf{u}_f$	p_f	ρ_s	$\mathbf{u}_s$	p_s	ϕ_s	t_f
RP2D1 [54]		$\gamma_f = 1.67$, $\pi_f = 0$			$\gamma_s = 1.4$, $\pi_s = 0$		$\delta = 10^3$, $\mu = 10^2$	
TR	1.5	(0.0,0.0)	2.0	2.0	(0.0,0.0)	2.0	0.8	0.15
TL	0.5	(0.0,0.0)	1.0	1.0	(0.0,0.0)	1.0	0.4	
BL	1.5	(0.0,0.0)	2.0	2.0	(0.0,0.0)	2.0	0.8	
BR	0.5	(0.0,0.0)	1.0	1.0	(0.0,0.0)	1.0	0.4	
RP2D2 [54]		$\gamma_f = 1.4$, $\pi_f = 0$			$\gamma_s = 3.0$, $\pi_s = 100$		$\delta = 0$, $\mu = 0$	
TR	1.0	(0.0,0.0)	1.0	1000.0	(0.0,0.0)	600.0	0.3	0.15
TL	1.5	(0.0,0.0)	2.0	800.0	(0.0,0.0)	500.0	0.4	
BL	1.0	(0.0,0.0)	1.0	1000.0	(0.0,0.0)	600.0	0.3	
BR	1.5	(0.0,0.0)	2.0	800.0	(0.0,0.0)	500.0	0.4	

Table 4
Numerical convergence results for the Taylor–Green vortex problem, for different Mach numbers M. The errors are given in L_1, L_2, and L_∞ norms, at the final time $t_f = 0.2$, for the single-phase and two-phase case.

M	L_1	$\mathcal{O}(\nabla\cdot\mathbf{u}=0)$	L_2	$\mathcal{O}(\nabla\cdot\mathbf{u}=0)$	L_∞	$\mathcal{O}(\nabla\cdot\mathbf{u}=0)$
	Single-phase					
10^{-1}	2.071400E−2	–	4.188600E−3	–	1.533700E−3	–
10^{-2}	4.964400E−4	1.62	1.026100E−4	1.61	3.603900E−5	1.63
10^{-3}	4.143400E−6	2.08	8.689600E−7	2.07	3.011100E−7	2.08
	Two-phase					
10^{-1}	2.9392E−02	–	5.5657E−03	–	1.9747E−03	–
10^{-2}	7.9792E−04	1.57	1.4673E−04	1.58	5.2690E−05	1.57
10^{-3}	4.1945E−06	2.28	8.7848E−07	2.22	3.0879E−07	2.23

4.5. Shock-bubble interaction

The shock-bubble interaction (SBI) problem [76] considers a planar shock wave in a liquid, interacting with a gas bubble, which leads to shock refraction and reflection. Such a problem finds many practical applications [77]. This test case allows us to show the ability of the SIFV-BN scheme to work also with multiple and interacting shocks. The computational domain is $\Omega = [-0.5, 3] \times [-0.75, 0.75]$, with a gas bubble of radius 0.25 located at $[0.5, 0.0]$, and governed by the ideal gas EOS, with $\gamma_f = 1.4$. The liquid phase is moving at high speed from left to right in the longitudinal direction, and is characterized by the stiffened gas EOS, with $\gamma_s = 3.0$ and $\pi_s = 100$. The domain is characterized by inflow conditions on the left boundary, slip-wall conditions on the other boundaries, and is discretized with $N_x \times N_y = 2048 \times 1024$ cells. In Fig. 14, several snapshots of the numerical solution are shown until the final time $t_f = 0.025$, both for the volume fraction and the liquid density. The shock waves are accurately captured, accounting for the several interactions with the wall boundaries on the top and bottom, and with one another. The results are qualitatively in very good agreement with those available in the literature [78]. This test case shows the ability of the numerical scheme to deal with highly complex multidimensional compressible flows.

4.6. Water-air mixture

To show the effectiveness of the numerical scheme also in the low Mach regime, a test involving water and air is conducted. The test is a Riemann problem inspired from Saurel et al. [19], and subsequently modified in Re and Abgrall [46]. The computational domain $\Omega = [-0.65, 0.65]$ is discretized with $N_x = 1000$ cells, with Dirichlet boundary conditions. The air is characterized by a density of $\rho_f = 1.2$, and follows the ideal gas EOS, with $\gamma_f = 1.4$. The water is characterized by density $\rho_s = 1050$, and follows the stiffened gas EOS, with $\gamma_s = 4.4$ and $\pi_s = 6.8 \times 10^8$. The two fluids are initially at rest, with uniform density, while pressure and volume fraction are discontinuous, with same values for both phases: $p = 10^6$ and $\phi = 0.3$ in the center of the domain, and $p = 10^5$ and $\phi = 0.7$ elsewhere. The air flow lies in the compressible regime and may show shock waves, while the water regime is incompressible. In this case, the definition of the interface velocity and pressure slightly modifies the water behavior. The version in Re and Abgrall [46] considers the definition (3b). The results are shown in Fig. 15 at the final time $t_f = 350\,\mu\text{s}$, showing an excellent agreement with the reference solution that has been computed resorting on the same TVD scheme used for the Riemann problems presented in Section 4.2. The Mach number of the liquid phase is of the order 10^{-3}, while the air flow reaches a Mach number of 1, thus involving a jump of four orders of magnitude within the same simulation. The SIFV-BN all Mach solver has proven to handle also this type of situations. The computational efficiency is compared against a fully explicit finite volume scheme, and the results are shown in Table 5. The SIFV-BN scheme is 4 times faster than the explicit method, which is particularly relevant for low Mach number flows like for this test case.

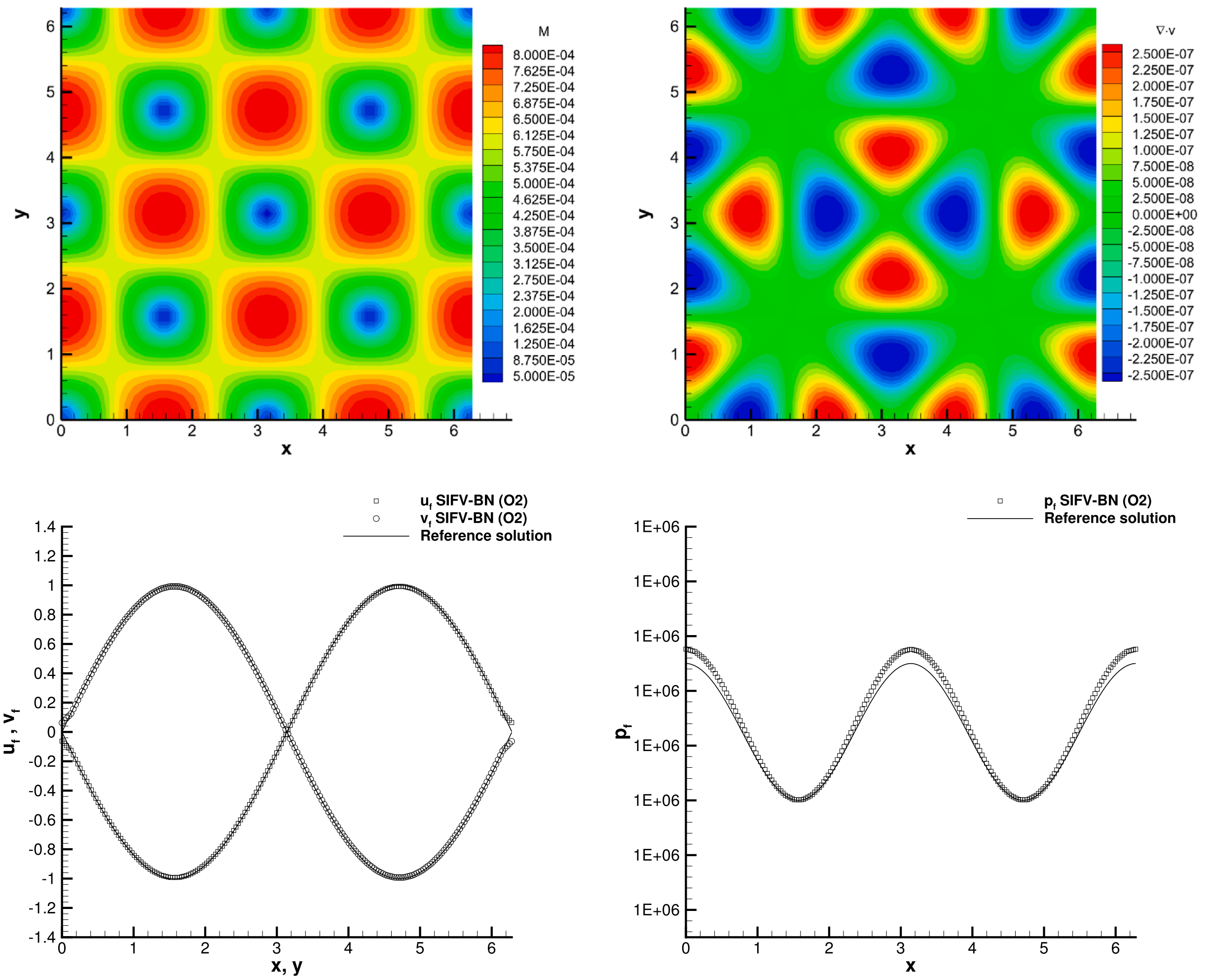

Fig. 13. Numerical solution of the two-dimensional Taylor–Green vortex for $M = 10^{-3}$ at the final time $t_f = 0.2$, using the SIFV-BN scheme. Top left: Mach number contours. Top right: mixture velocity divergence contours. Bottom left: 1D cuts along the $x-$ and the $y-$axis, and comparison with the reference solution for the fluid mixture velocity components u and v. Bottom right: 1D cut along the $x-$axis for the fluid mixture pressure p, and comparison to the reference solution.

Table 5
Computational time of the SIFV-BN scheme compared against the explicit Finite Volume scheme for the test cases presented in Sections 4.6 and 4.2. Times are measured in seconds, and the simulations have been run on a 12th Gen Intel(R) Core(TM) i7-1255U 1.70 GHz processor on 4 CPU. The ratio is computed as $\beta =$ Explicit FV/SIFV-BN.

Test case	SIFV-BN	Explicit FV	β
Explosion problem	131.46	529.22	4.03
Water-air mixture	14194.19	614122.20	43.27

Furthermore, this problem can be extended to three space dimensions, on a cubic computational domain $\Omega = [-0.65, 0.65]^3$. Two different simulations are run until the final time $t_f = 300\,\mu\text{s}$, to show mesh convergence in the incompressible limit of the liquid phase: the first one involves 7’077’888 cells, while the second one is concerned with 16’777’216 cubic control volumes. The results are plot in Fig. 16, where we can see that no spurious oscillations arise, and that the symmetry of the numerical solution is well preserved.

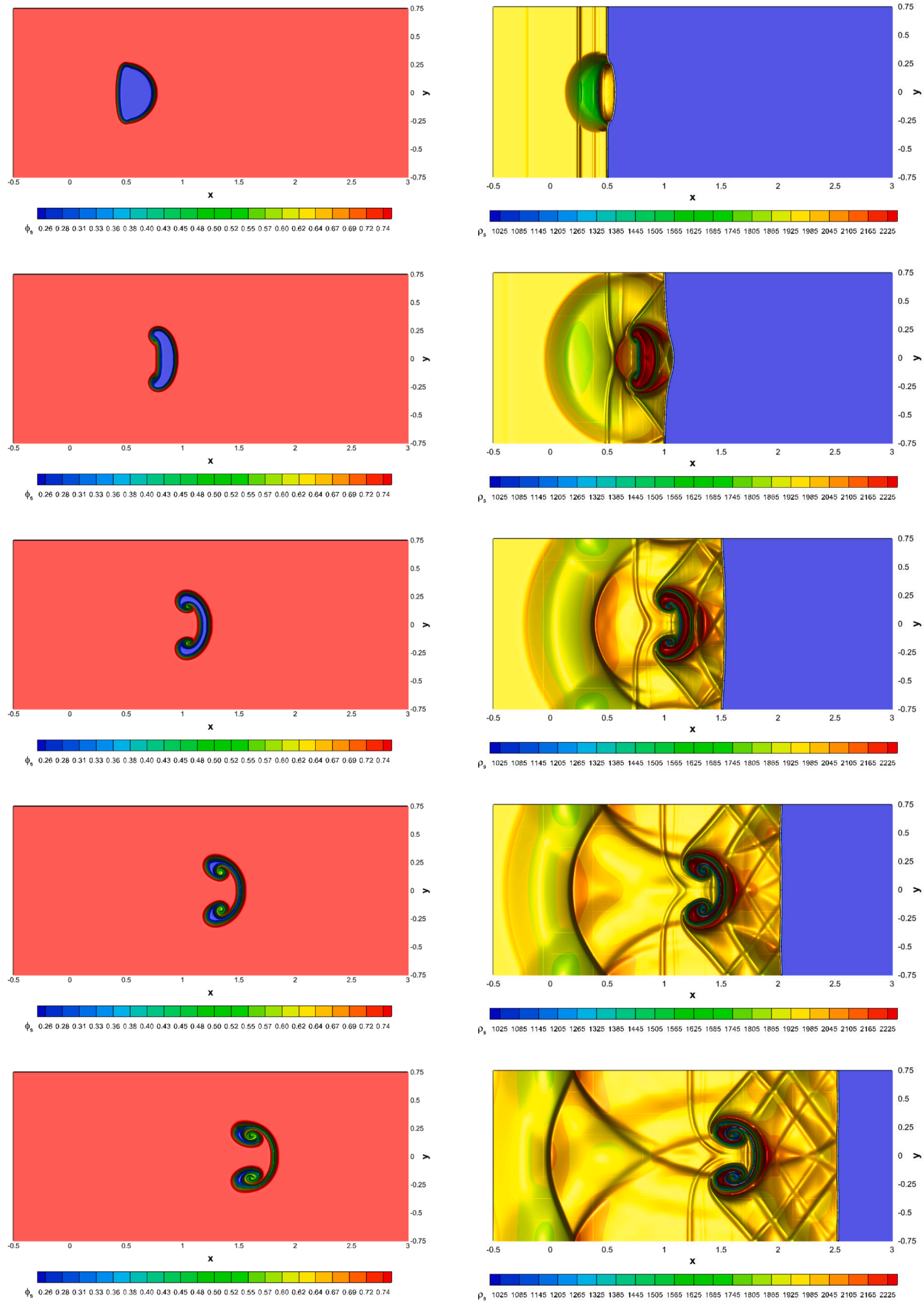

Fig. 14. Shock-bubble interaction problem at times $t = 0.005$, $t = 0.010$, $t = 0.015$, $t = 0.020$ and $t = 0.025$ (from top to bottom). Numerical results for volume fraction (left) and solid phase density (right) depicted with 21 equidistant contour lines.

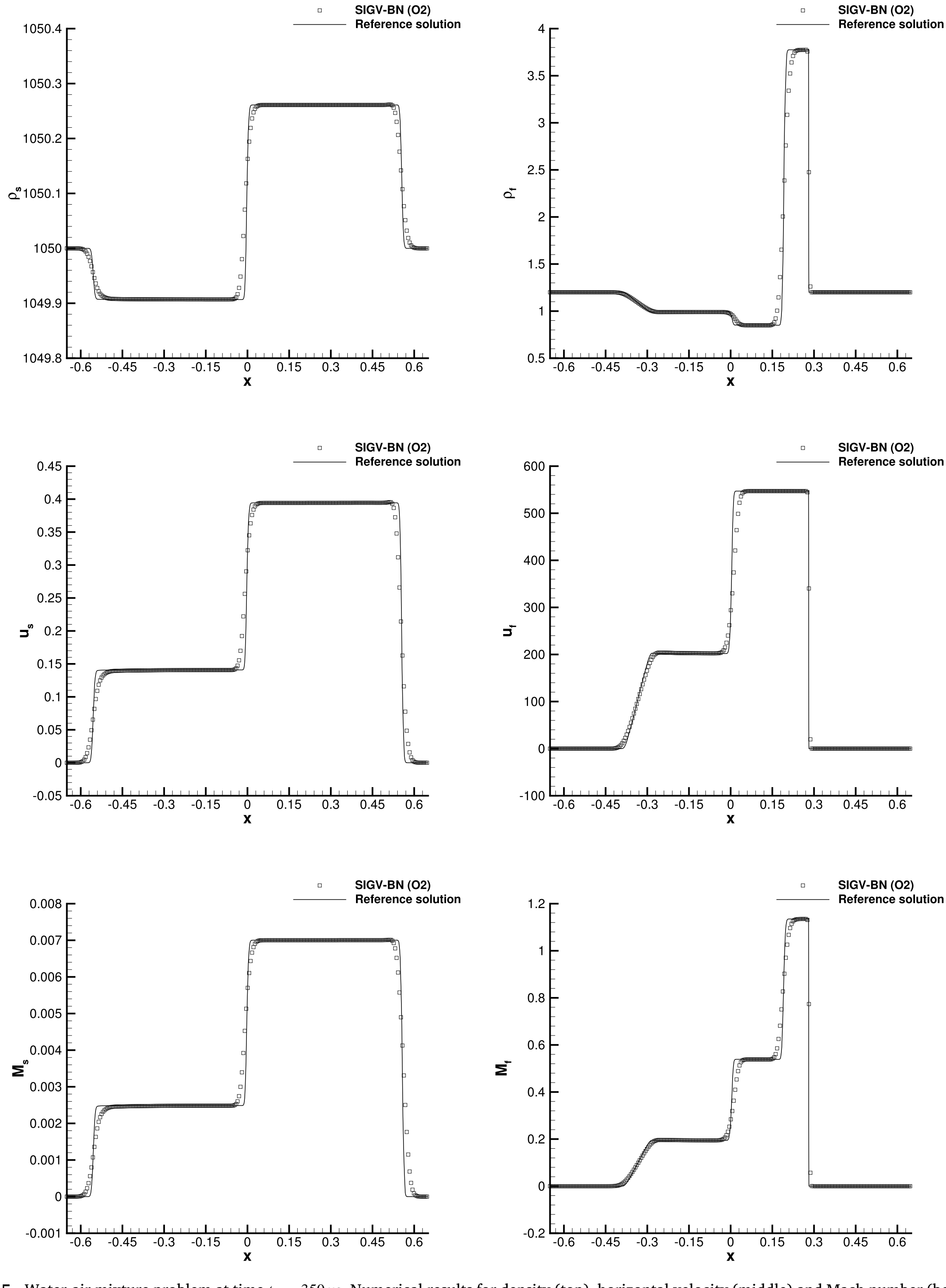

Fig. 15. Water-air mixture problem at time $t_f = 350\,\mu\mathrm{s}$. Numerical results for density (top), horizontal velocity (middle) and Mach number (bottom) compared against the reference solution. Left: solid phase. Right: fluid phase.

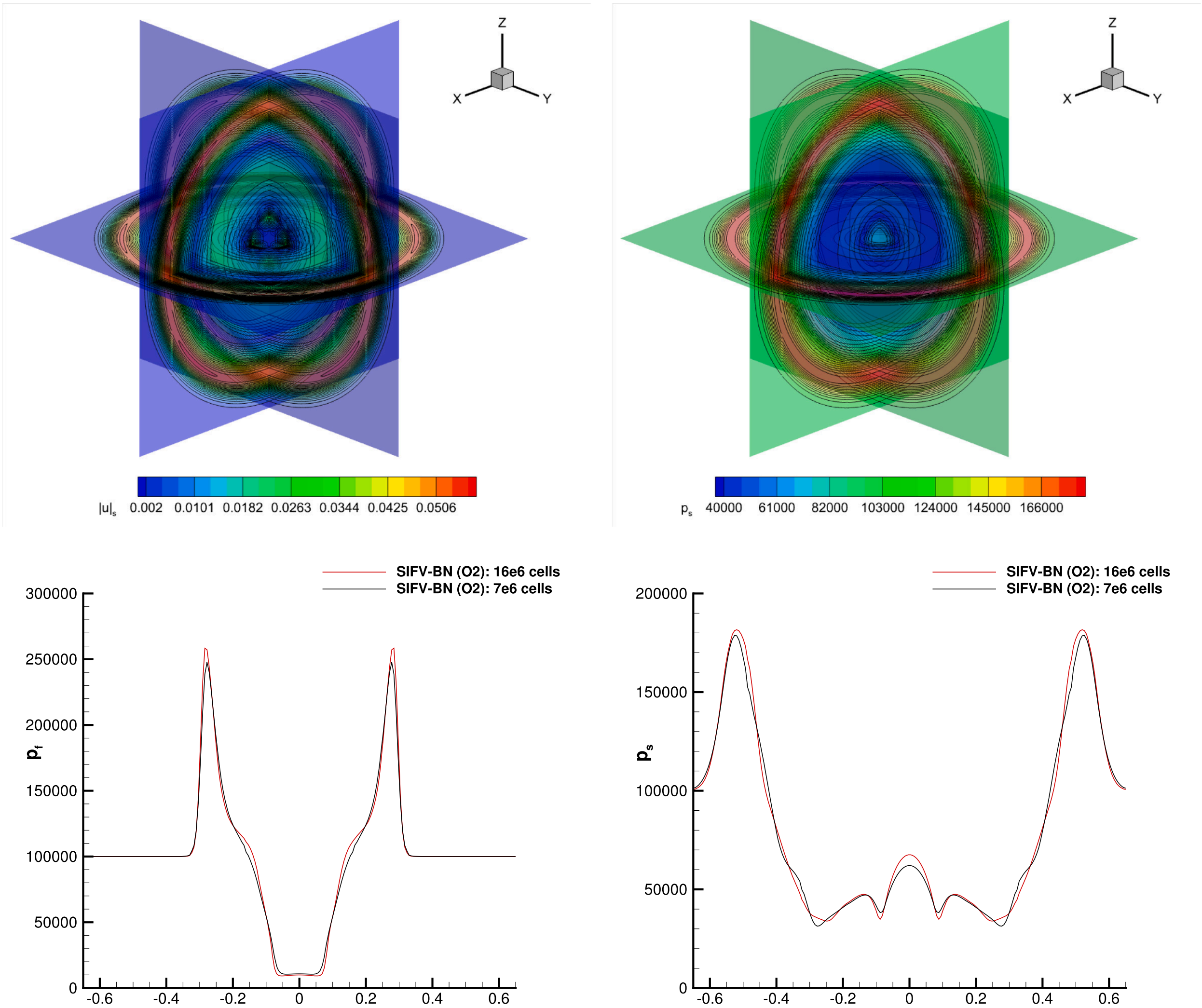

Fig. 16. Water-air mixture problem in 3D at time $t = 300\,\mu\text{s}$. Top: numerical distribution of the liquid velocity magnitude $|\mathbf{u}_s|$ (left), and pressure p_s (right), along the planes $x = 0$, $y = 0$, $z = 0$. Bottom: air (left) and water (right) pressure solution obtained with two different computational meshes.

5. Conclusions

In this paper, a semi-implicit numerical scheme is presented for the solution of the Baer–Nunziato model for compressible two-phase flows, to deal with all Mach number regimes. The pressure fluxes and the potentially stiff source terms are discretized implicitly with central finite differences, while shock capturing properties are guaranteed by an explicit finite volume method for the remaining convective terms. In this way, the new scheme avoids an excessive restriction on the time step, that is only based on the mean flow velocity. The numerical approximation of the non-conservative products relies on physical considerations, which ultimately allows the scheme to exactly preserve moving equilibrium solutions of the governing equations. The scheme is proven to be second order in time and in space, with a judicious discretization of the numerical viscosity. To identify the different Mach number regimes, the non-dimensional equations are retrieved using a Mach number common to both phases, hence dealing with the mixture flow. The asymptotic preserving property of the scheme in the low Mach number limit model for the mixture is also demonstrated. The novel solver shows accuracy and robustness both in the compressible regime and approaching the incompressible limit.

Future works will consider the definition of different Mach numbers for each phase, to design asymptotic preserving schemes. More physics will also be tackled by including the viscosity terms and the gravity terms, yielding to the identification of different Reynolds and Froude numbers. This will allow us to deal with more complex models, such as the simulation of magma flows in a volcanic conduit. Finally, since primitive variables are physically measurable in real world scenarios, we plan to discretize the Baer–Nunziato model written in fully non-conservative form, while ensuring conservation properties.

CRediT authorship contribution statement

Beatrice Battisti: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation; **Walter Boscheri:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Walter Boscheri reports financial support was provided by French National Research Agency and by Italian Ministry of University and Research. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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